

Supplementary Information for

Strategies for climate-resilient global wind and solar power systems

Dongsheng Zheng^{1,#}, Xizhe Yan^{1,#}, Dan Tong^{1,*}, Steven J. Davis², Ken Caldeira^{2,3,4}, Yuanyuan Lin¹, Yaqin Guo¹, Jingyun Li¹, Peng Wang¹, Liying Ping¹, Shijie Feng¹, Yang Liu⁶, Jing Cheng², Deliang Chen¹, Kebin He^{5,6}, Qiang Zhang^{1,6,*}

¹Ministry of Education Key Laboratory for Earth System Modeling, Department of Earth System Science, Tsinghua University, Beijing, 100084, China.

²Department of Earth System Science, Stanford University, Stanford, California, USA.

³Gates Ventures, Kirkland, Washington, USA.

⁴Department of Global Ecology, Carnegie Institution for Science, Stanford, California, USA.

⁵State Key Joint Laboratory of Environment Simulation and Pollution Control, School of Environment, Tsinghua University, Beijing, 100084, China.

⁶Institute for Carbon Neutrality, Tsinghua University, Beijing, 100084, China.

These authors contributed equally to this work.

*corresponding authors: dantong@tsinghua.edu.cn; qiangzhang@tsinghua.edu.cn

Table of Content

Supplementary Text 1. Model validation

Supplementary Text 2. Uncertainty analyses

- 2.1 Climate-driven electricity demand
- 2.2 Cost-optimized wind and solar dispatch
- 2.3 Electricity transmission
- 2.4 Hourly cost estimation during extreme periods
- 2.5 Mitigation strategy

Supplementary Text 3. Supplementary results

- 3.1 Climate-driven supply and demand
- 3.2 Changes in hourly costs
- 3.3 Drivers of hourly cost changes
- 3.4 Strategies to mitigate hourly cost increases
- 3.5 Policy discussion

Supplementary Figures

Supplementary Tables

Supplementary References

Supplementary Text 1. Model validation

Previous studies typically validate electricity dispatch models based upon system costs and installed generation capacities¹⁻³. This study also focuses on endogenous system costs and technology-specific installed capacities. In specific, we employ historical electricity demand, cost data, and climatic inputs from the ERA5 reanalysis dataset during 2016–2020. Based on these data, we reconstruct power system costs and installed capacities in the near-current periods, enabling a robust comparison between model outputs and real-world benchmarks.

To validate the system cost, we combine the levelized cost of energy (LCOE) for different technologies (Supplementary Table 21) with their respective generation shares in 2020 (Supplementary Table 17) to estimate current hourly electricity generation costs. The results show that the dispatch optimization model exhibits a strong agreement with the empirical cost estimates and effectively captures country-level differences, particularly for larger nations (Supplementary Fig. 19). There are 22 out of 26 countries showing a cost deviation within $\pm 10\%$, with large economies such as the U.S. and India exhibiting differences of less than $\pm 5\%$. Larger discrepancies observed in restively small countries may be partly attributed to data limitations, as localized cost information is unavailable. In such cases, regional or global averages were used, potentially introducing bias. Overall, these results suggest that the dispatch model provides a robust and reliable representation of national-level power system costs.

Regarding the validation of installed capacities, we collect 2020 data for wind, solar, hydro, nuclear, and thermal power from the International Renewable Energy Agency (IRENA; <https://pxweb.irena.org/pxweb/en/IRENASTAT/>). Overall, our dispatch model outputs are largely consistent with the reported capacities, despite potential overestimates and underestimates (Supplementary Fig. 18). Most technologies are reproduced with reasonable accuracy. For wind and solar, the model slightly overestimates capacities in some countries with large and rapidly expanding renewable sectors, such as China and the U.S. These discrepancies likely reflect a combination of climate uncertainties and potential gaps or delays in official reports. Similar uncertainties in capacity modeling have also been documented in other dispatch models, such as PyPSA^{1,2}. Nevertheless, the overall installed capacity structure produced by our model remains highly consistent with observed data.

Supplementary Text 2. Uncertainty analyses

2.1 Climate-driven electricity demand

To assess the uncertainties arising from climate sensitive electricity demand, we performed sensitivity analyses focusing on the temperature–demand response parameters and the proportion of climate-sensitive electricity demand. For the temperature–demand response parameters, we followed uncertainty ranges reported in previous studies^{4,5} and designed two representative tests. The aggressive test assumes a lower cooling threshold of 18.8°C but a higher heating threshold of 15.2°C, with a response coefficient of 5.0% for cooling and 3.0% for heating. The conservative test assumes a higher cooling threshold of 20.4°C but a lower heating threshold of 9.7°C, with a response coefficient of 3.0% for cooling and 2.0% for heating. Additionally, we varied the shares of temperature-dependent residential electricity demand by 50% and 70%. Across all tests, changes in hourly cost during extreme periods remain comparable to the baseline and are even aggravated under the aggressive test (Supplementary Fig. 12), suggesting that our core findings are not sensitive to these key input assumptions.

2.2 Cost-optimized wind and solar dispatch

To evaluate the uncertainty of wind and solar supply in the cost-minimized optimization, we performed sensitivity tests by adjusting wind and solar supply targets, wind/solar generation ratios, and key assumptions related to dispatch patterns (i.e., carbon capture, utilization, and storage (CCUS) consideration, storage costs, thermal and hydro supply share, as well as nuclear and hydro flexibility). In specific, the effects of renewable supply targets are examined using aggressive (90th percentile) and conservative (10th percentile) projections by Integrated Assessment Model (IAM; Supplementary Table 22). The influence of different wind/solar generation ratios is tested based on IAM projections by 2060 (Supplementary Table 23), rather than treating them as an endogenous variable in the base model. For CCUS, we add new decision variables to represent this technology in the base model constrained by the IAM-projected penetration rates by 2060 (Supplementary Table 24). Regarding storage cost, we tested two alternative cost levels by increasing and decreasing both fixed and variable costs by 10% relative to the baseline. For thermal power, we test a reduced thermal supply share corresponding to the 10th percentile of IAM projections (Supplementary Table 25). In the hydropower test, we increase the hydro share by 10% relative to the base case, while assuming fixed output profile aligned with electricity demand. For nuclear power, we relax the baseline constant 100% output limit and test an 80% minimum output. Across all tests, increases in hourly costs during extreme periods remain notably consistent—even more significant than the base case (Supplementary Fig. 12)—indicating that these tests do not alter our main conclusions.

2.3 Electricity transmission

In addition, to evaluate the influence of spatial simplification, we develop a nested dispatch model that incorporates existing alternating current (AC) and direct current (DC) transmission

topography and capacity (Supplementary Tables 12–13). The simulation is conducted at 3-hour resolution rather than 8760-hour simulation to reduce computational complexity, while still capturing seasonal and diurnal variations in wind-solar supply and electricity demand. We apply this nested model to assess the effects of subnational interconnection across seven major countries (i.e., the U.S., China, India, Germany, Brazil, South Africa, and Australia), and to examine the impact of international interconnection across 38 European countries. Results show that the nested model yields hourly cost change generally consistent with the base model (Supplementary Figs. 10–11).

2.4 Hourly cost estimation during extreme periods

For the evaluation of hourly cost change during extreme periods, we examine the influence of alternative approaches to hourly cost decomposition and the identification of extreme periods. Regarding cost decomposition, we conduct three sensitivity tests. In the first test (i.e., *CF10th* method), we apply a method that calculates the “hours of effective utilization per unit of installed capacity” excluding hours when wind or solar capacity factors fall below the 10th percentile (excluding zero values). The second test (i.e., *CFmax* method) adopts the concept of “output revenue per unit of installed capacity” based on the premise that higher capacity factors typically yield greater electricity revenues, thereby justifying a smaller allocation of fixed costs. In this approach, we define the output revenue as the difference between the annual maximum hourly capacity factor and the actual hourly capacity factor. The fixed costs are then allocated using hourly weighting factors derived from sorting these values in an ascending order. In the third test (i.e., *CFmax_weight* method), we exclude hours with zero wind or solar capacity factors and directly apply the hourly output revenue as weighting factors for fixed cost allocation. For the identification of extreme periods, the baseline case uses the 90th percentile threshold, while an additional sensitivity test applies a more stringent 95th percentile threshold. Results from these sensitivity tests confirm that the observed changes in hourly costs during extreme periods remain consistently significant, demonstrating the robustness of our key findings across different cost post-processing approaches (Supplementary Fig. 12).

2.5 Mitigation strategy

In the strategy design phase, we examine uncertainties in key parameter assumptions under both conservative and aggressive tests (Supplementary Table 19). Specifically, for demand response, a peak response coefficient of 7% is assumed under the aggressive test and 3% under the conservative one^{6,7}. For expanded interconnection, the maximum shares of electricity demand met by electricity imports are set at 10% in the aggressive case and 3% in the conservative case^{8,9}. Regarding enhanced storage, discharge/charge efficiency is assumed to reach 99.5% for short-duration storage and 99% for long-duration storage under the aggressive test, whereas in the conservative test short-duration storage and long-duration storage assume 97% and 93%, respectively^{10,11}. For improved thermal flexibility, the minimum output levels

are set at 25% for coal, 20% for oil, and 15% for gas in the aggressive test, compared to 35%, 30%, and 25% in the conservative test^{12,13}. Our results indicate that aggressive strategies can enhance the system's ability to mitigate climate-related hourly cost increases, while conservative strategies show more moderate effectiveness (Supplementary Figs. 14–15). However, both results remain comparable to the base case. These findings underscore the need for policy-makers to prioritize ambitious system designs that enhance power system resilience to address climate risks.

Supplementary Text 3. Supplementary results

3.1 Climate-driven supply and demand

The global decline in wind speeds remains ongoing debate¹⁴⁻¹⁶, but it is generally attributed to global atmospheric circulation patterns^{17,18}. In specific, climate warming tends to be more pronounced at high latitudes¹⁹, leading to a reduced latitudinal pressure gradient. It may be a key factor driving the global downtrend in wind speeds, particularly in high- and middle-latitude regions in the study (Figs. 1a–b). We also observe an increase in wind speeds in some low-latitude regions, particularly in southern hemisphere such as Australia, Brazil, and the Congo (Figs. 1a–b). This increase is likely driven by enhanced land-ocean temperature contrasts, known as land amplification¹⁵. In brief, a lower specific heat capacity of land surfaces allows to a more pronounced warming compared to the ocean, which increases the pressure gradient between land and ocean, thereby strengthening wind speeds in these regions.

Variability in electricity demand is also largely dependent on latitudes. In specific, middle- and high-latitude regions experience a greater seasonal and diurnal demand variability than low-latitude regions^{5,20}, likely driven by more pronounced temperature fluctuations between seasons and between day and night²¹. Specifically, these regions typically experience high temperatures in summer and low temperatures in winter, leading to large cooling/heating demand during the two seasons. In contrast, spring and autumn are generally milder, resulting in a lower electricity demand. As a result, electricity demand in middle- and high-latitude countries, such as Canada and Kazakhstan, likely exhibits a clear seasonal variability²⁰. In comparison, low-latitude regions tend to have more stable temperatures throughout the year, leading to smaller seasonal demand fluctuations²⁰. Additionally, middle- and high-latitude regions often experience a greater diurnal temperature variation²⁰, possibly due to a lower cloud cover and a more pronounced atmospheric circulation, contributing to a higher diurnal demand variability compared to low-latitude regions.

3.2 Changes in hourly system costs

Power systems may face substantial yet often underestimated economic cost risks under more severe climate conditions. As illustrated in the Extended Data Fig. 9, most countries experience significant increases in hourly costs under the dispatch model simulation with the largest hourly cost increases across climate model ensembles, even when compared to the 95th percentile regardless of extreme and normal periods. Such surge in hourly costs is particularly pronounced in Asian and African countries. For instance, Tajikistan's hourly costs escalate from 23.7% to around 60%, and Botswana's hourly costs rise from 8.5% to over 25%. These findings suggest that, in the face of climate extremes, these nations may incur substantially higher costs to meet the same electricity generation share, potentially exacerbating financial pressures on their national budgets. Accordingly, preemptive and adaptive strategies may be critical to safeguard future power system security.

Given that future power system planning may place a greater emphasis on the financial stress associated with the most severe climate change impacts, we quantify the additional economic costs expressed as Gross Domestic Production (GDP) loss (Supplementary Tables 4–5). The average annual GDP loss is calculated as a percentage of annual additional climate-related costs, relative to the IAM-projections by 2060 (<https://data.ece.iiasa.ac.at/ar6/>). We find that GDP loss increases significantly under more severe climate impacts, compared to the average level, especially for small developing countries. For instance, in Tajikistan, under the SSP126 scenario, the average annual GDP loss is only 1.6%, while it could rise to 3.7% under the 95th percentile of dispatch model simulations with climate model ensembles. Additionally, despite some countries experiencing limited GDP loss, they are not usually exempt from substantial increases in annual total system costs. For example, in the U.S., although the average annual GDP loss is only 0.04%, it is projected to experience a severe growth in annual total system cost (\$15.7 billion) under the SSP126 scenario. Furthermore, it could potentially reach as high as \$38.1 billion under the 95th percentile of dispatch model simulations with climate model ensembles. These results emphasize the necessity for policymakers—both in developing and developed countries—to integrate such extremes into long-term power system resilience planning.

3.3 Drivers of hourly cost changes

Changes in system costs result from the interplay of multiple factors, making it challenging to attribute the impacts to individual drivers. This study considers two standard climate scenarios (i.e., SSP126 and SSP245), reflecting different levels of challenges related to climate change, energy transition pathway (i.e., wind/solar energy expansion, a key model configuration in this study), and economic development. The SSP126 and SSP245 share common but distinct risks related to wind-solar penetration and climate change levels. In specific, SSP245 represents a much higher level of temperature rise, which likely allows a much higher demand variability due to higher cooling demands, thus possibly leading to a severer risk in hourly cost growth. Given that a certain SSP corresponds to a specific set of radiative forcing, the changes in hourly costs between different SSPs are not directly comparable. Therefore, these combined effects of multiple factors on changes in hourly costs may be difficult to isolate. As an alternative, we could compare the differences between wind-solar supply changes and demand variability across varied climate scenarios. For example, the increase in hourly costs for Madagascar under SSP245 may be attributed to increases in demand variability, as the compound low wind-solar event duration is actually reduced compared to SSP126 (Figs. 1–3).

The detailed analysis of seasonal and diurnal distributions of extreme hours is crucial for understanding their underlying drivers. We find that the seasonal distribution varies significantly across regions, likely depending on their climatic conditions (Supplementary Fig.

4). For example, China experiences pronounced summer peaks (June–August) due to a higher cooling demand and low wind power generation. However, the Russia shows dual peaks—one in winter, likely associated with solar droughts, and another in summer, likely driven by a lower wind supply. As for diurnal variations, extreme hours consistently show two distinct global peaks (Supplementary Fig. 4): an early-morning peak (00:00–06:00), resulting from the absence of solar generation, and a more pronounced evening peak (17:00–20:00), driven by the sharp decline in solar power output but increased residential electricity demand. The temporal pattern of extreme hours may guide the development of targeted mitigation strategies. Regions with pronounced seasonal demand peaks may necessitate the deployment of long-duration storage solutions like hydrogen and pumped hydro. Conversely, the diurnal demand surge likely underscores the importance of peak-shifting strategies such as dynamic electricity pricing mechanisms.

With comparison to extreme periods, normal periods also have important nuances in the drivers of hourly cost increases across countries. During normal periods, while storage systems represent an important determinant of hourly cost increases (as seen in Brazil and Egypt), our findings reveal that in most key countries, the rise in hourly costs is primarily driven by greater investments in wind and solar capacities needed to maintain the same generation share. For instance, under the SSP126 scenario, in U.S. solar-dominated renewable power system (Supplementary Fig. 7), a 0.13 increase in compound low wind-solar event duration and a 0.03 increase in demand variability leads to a 3.3% increase in hourly costs due to the rising solar power capacity cost (Figs. 1, 3). Similarly, in wind-dominated Japan, a 0.01 decrease in wind speed likely results in a 0.1% cost increase, mainly attributed to a higher wind power capacity expense (Figs. 1, 3).

The observed changes in hourly costs are closely linked to variations in the installed capacities of flexible energy sources, including thermal power, hydropower, and energy storage. To better understand this relationship, we assess the additional flexible energy capacity requirements and compare them with the near current investment levels observed in 2020. (<https://pxweb.irena.org/pxweb/en/IRENASTAT/>). We find that the primary contributor to the rising hourly costs may be the expansion of storage capacities (Supplementary Fig. 6). For instance, in China, a 5.1% increase in hourly costs during extreme periods are partly driven by an additional 111.03 GW long-duration storage, representing over 300% of the near current pumped hydro storage capacity. This impact is even more pronounced in countries with limited storage infrastructure like France (i.e., over 470%). Additionally, there are also substantial additional capacity requirements for thermal power and hydropower, due to their potential ability to provide backup resources. These findings underscore the importance of prioritizing flexible energy technologies in future energy system planning, as a cornerstone for building climate-resilient power systems.

3.4 Strategies to mitigate hourly system cost increases

Evaluating the effectiveness of mitigation strategies requires an explicit consideration of the more severe climate conditions, as the average outcomes may substantially overestimate their performances compared to truly catastrophic conditions. Therefore, we compare the effectiveness of mitigation measures in addressing both the top 5% of events with the largest cost increases and the event with the largest cost increase (Supplementary Fig. 9). The results reveal a significant decline in the likelihood of successfully mitigating cost risks when considering the absolutely worst case. For example, Russia, during normal periods, there is only a 30.3% likelihood that investments in resilience will reduce costs in the top 5% of events with the largest cost increases, while this likelihood drops to 25.1% in the event with the largest cost increase (Supplementary Fig. 9). Notably, this gap may be even more significant when incorporating a larger number of measures in some countries with substantial hydro power like Kazakhstan. This finding underscores that policy frameworks based upon near-extreme conditions may risk underpreparing for truly catastrophic outcomes. Designing flexible and robust strategies that remain effective under both common and most extremes will be crucial for enhancing the resilience of future power systems.

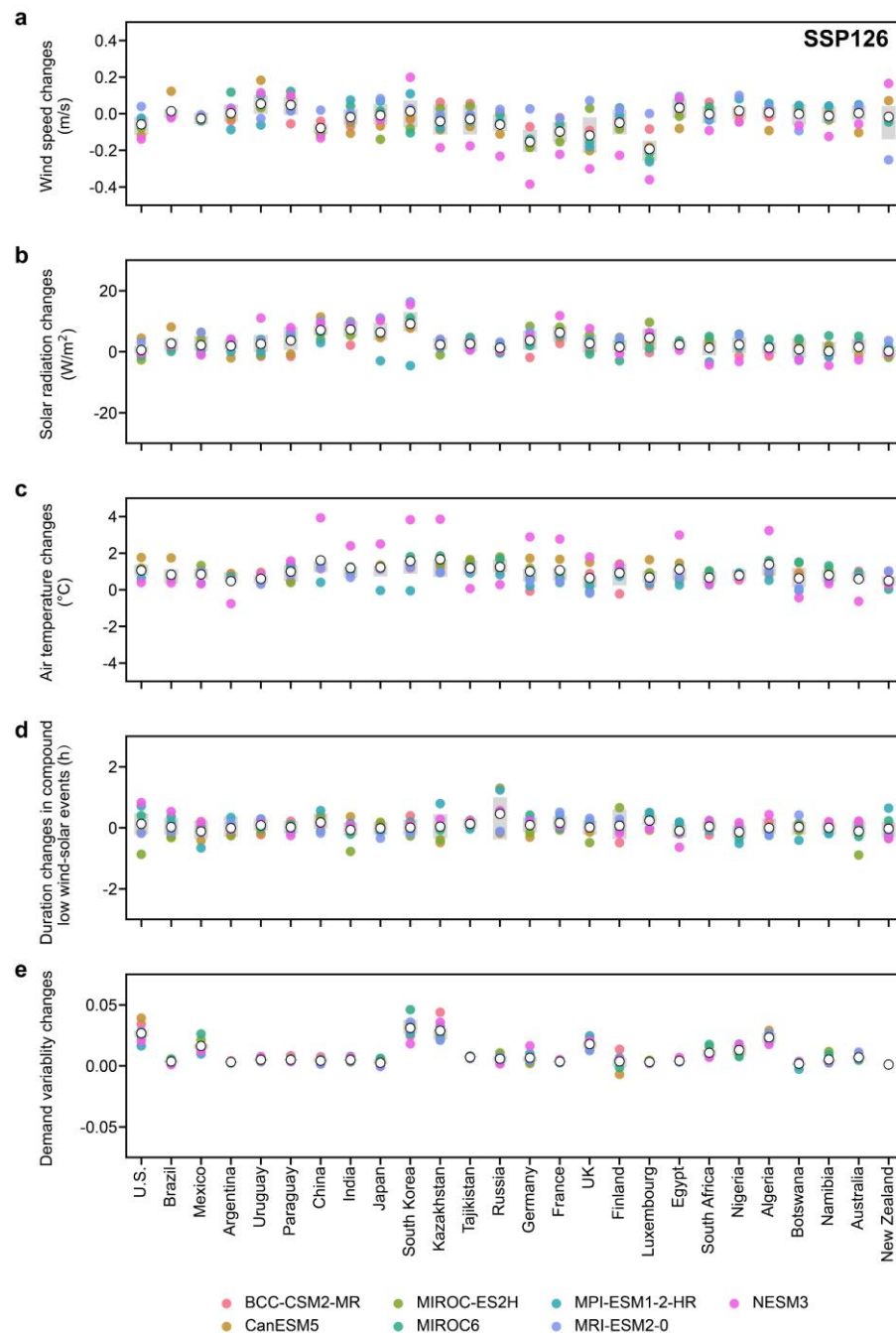
3.5 Policy discussion

Our analysis of hourly cost decomposition for power systems carries important policy implications on real-time electricity pricing. Flexible electricity pricing mechanisms inherently reflect the tensions arising from supply and demand mismatches. In European electricity markets, such mechanisms have been pioneered to mitigate these imbalances²², thereby enhancing renewable energy expansion and market efficiency. Specifically, electricity prices under these mechanisms fluctuate in real time based on the wholesale market, a system known as Real-Time Pricing (RTP), which updates electricity prices at hourly or shorter intervals²³⁻²⁵. Alternatively, Time-of-Use Pricing (TOU) sets different rates for specific periods of the day, such as the Peak/Off-Peak Pricing model widely adopted in Germany and France²⁶⁻²⁹. Another approach, Critical Peak Pricing (CPP), temporarily raises electricity prices during periods of grid stress—such as peak demand hours or extreme weather events—to incentivize demand reduction³⁰. These high-cost periods align closely with the extreme hours identified in our study. By simulating hourly costs under extreme conditions, policymakers can anticipate severe supply-demand mismatches and adjust electricity prices accordingly.

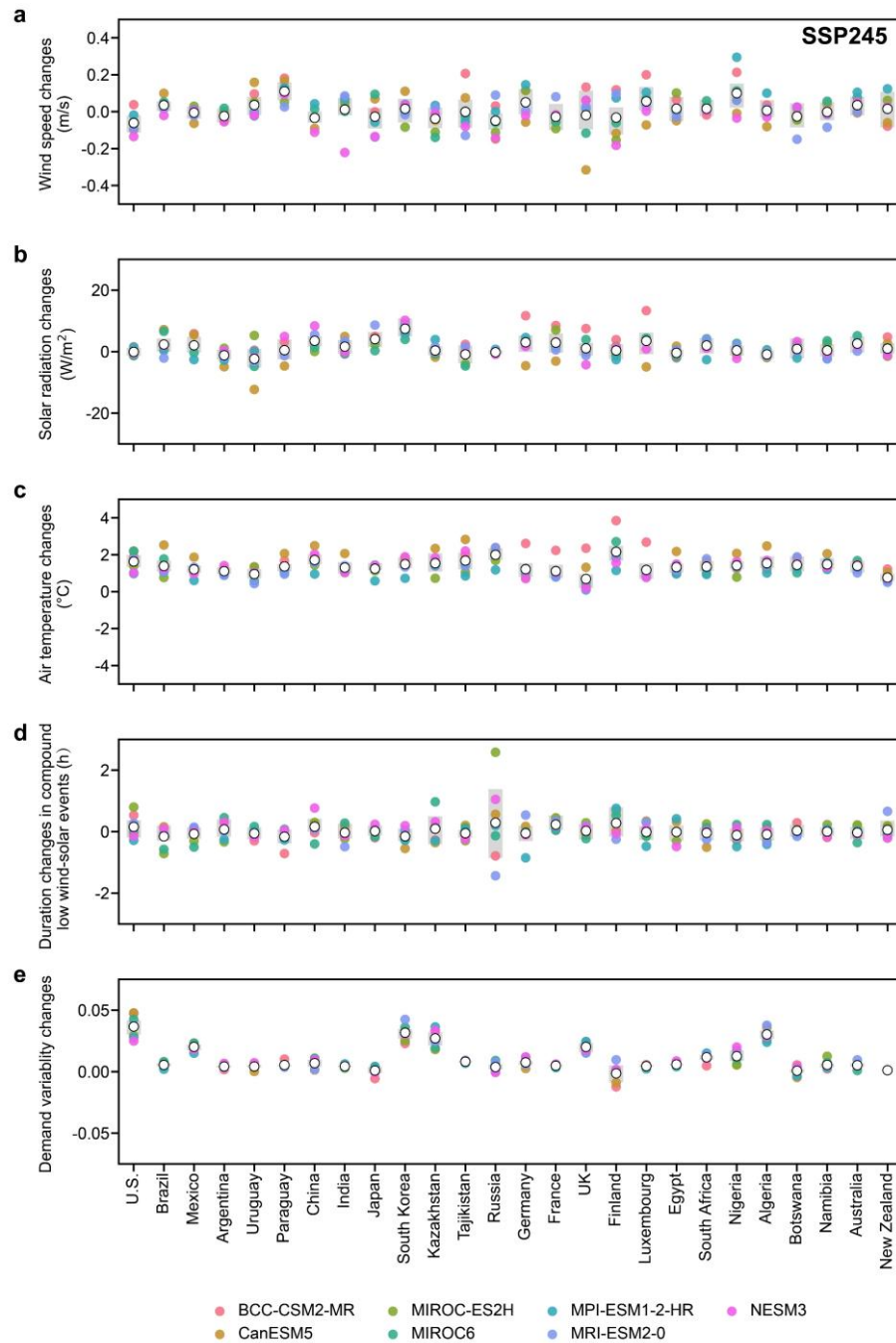
This study also offers important policy insights for managing systemic risks in interconnected power systems when considering synchronized renewable shortages. In power systems with high wind-solar supply shares, international transmission remains at risk of synchronized power shortages. Particularly, in Europe, nearly all countries experience extended periods of low wind capacity factors during June, July, and August (Supplementary Fig. 16). The primary challenge lies in their tendency to undermine the effectiveness of cross-border electricity trade as a

balancing mechanism, potentially driving up electricity prices and increasing reliance on fossil fuels. One approach to minimize the adverse effects of such synchronized risks is diversifying energy mix by integrating complementary renewable sources, such as hydropower and geothermal power. Additionally, enhancing storage efficiency alongside demand-side management and sector coupling, can provide crucial backup during renewable droughts and help reduce cascading failures. Furthermore, better forecasting techniques and coordinated regional energy system planning can support more effective anticipation and management of synchronized renewable shortage risks.

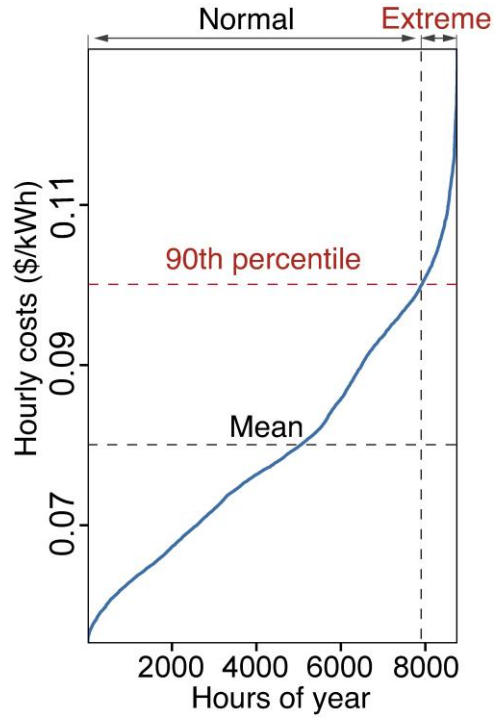
Supplementary Figures



Supplementary Figure 1 | Changes in key climate variables and their impacts on the wind-solar supply and electricity demand in 26 major countries projected by different climate models under the SSP126 scenario. Average changes in near-surface wind speeds (**a**), solar radiation (**b**), air temperature (**c**), durations of compound low wind-solar events (**d**), and total demand variability (**e**). In each case, the grey bars represent the 33rd to 67th percentile ranges across multi-model ensembles; the colored scatters indicate the average values of each single-model ensemble, while the white scatters represent the average values across all models. The change represents a comparison between future climate (2056–2060) and current climate (2016–2020).

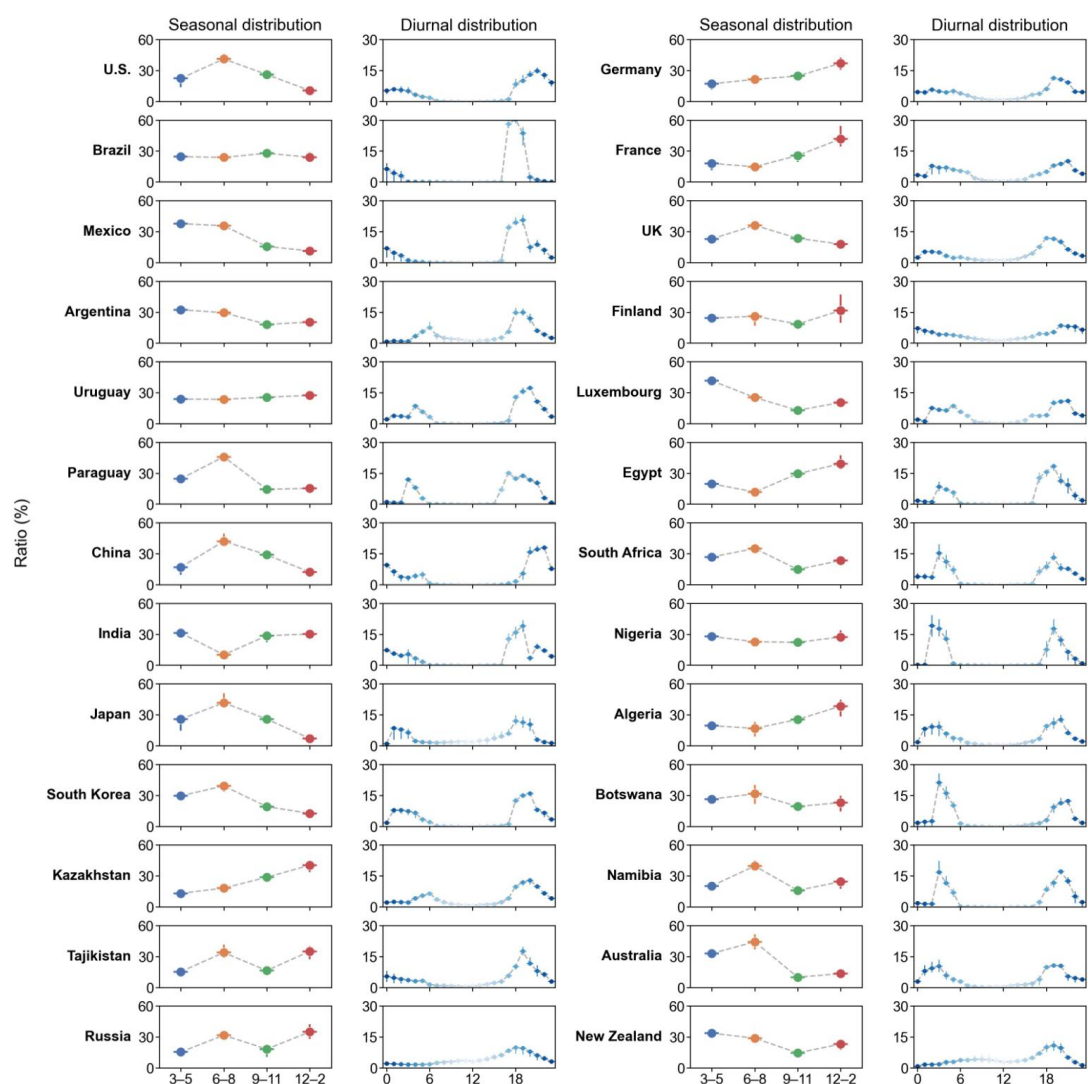


Supplementary Figure 2 | Changes in key climate variables and their impacts on the wind-solar supply and electricity demand in 26 major countries projected by different climate models under the SSP245 scenario. Average changes in near-surface wind speeds (**a**), solar radiation (**b**), air temperature (**c**), durations of compound low wind-solar events (**d**), and total demand variability (**e**). In each case, the grey bars represent the 33rd to 67th percentile ranges across multi-model ensembles; the colored scatters indicate the average values of each single-model ensemble, while the white scatters represent the average values across all models. The change represents a comparison between future climate (2056–2060) and current climate (2016–2020).

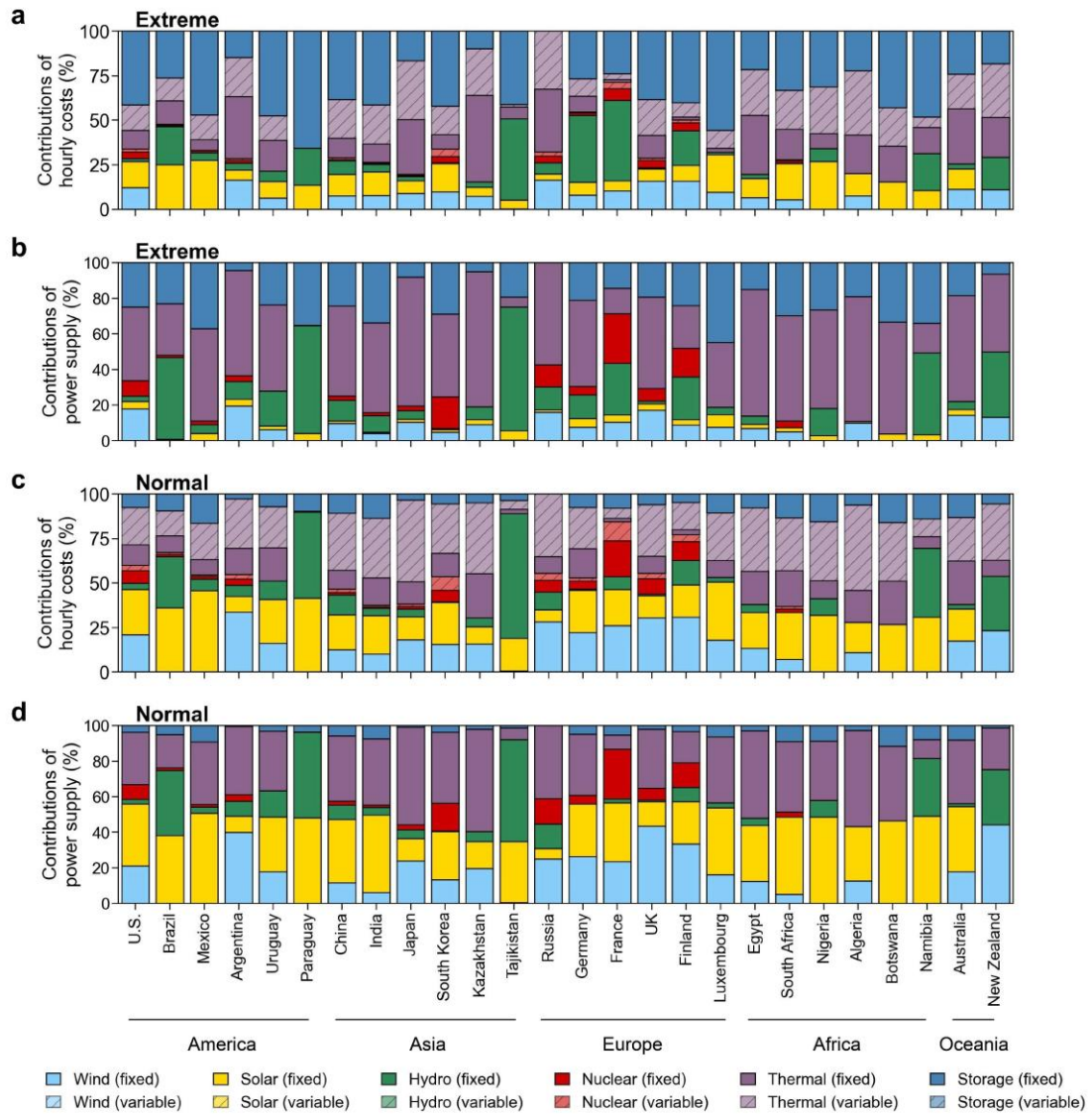


Supplementary Figure 3 | Schematic of hourly costs during extreme and normal periods.

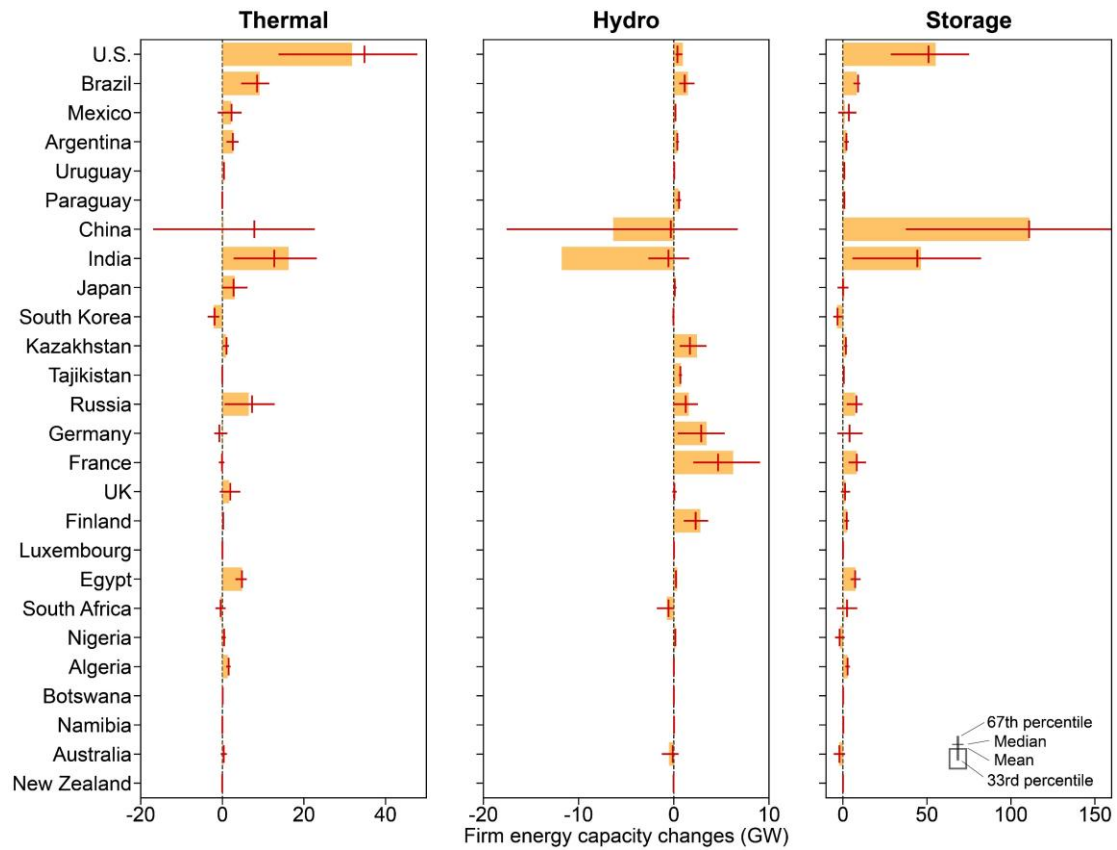
The hourly cost is ranked for all 8,760 hours of the year, with hours in the upper decile of hourly costs (i.e., the most-costly 10% of hours) classified as extreme periods, and the remaining as normal periods.



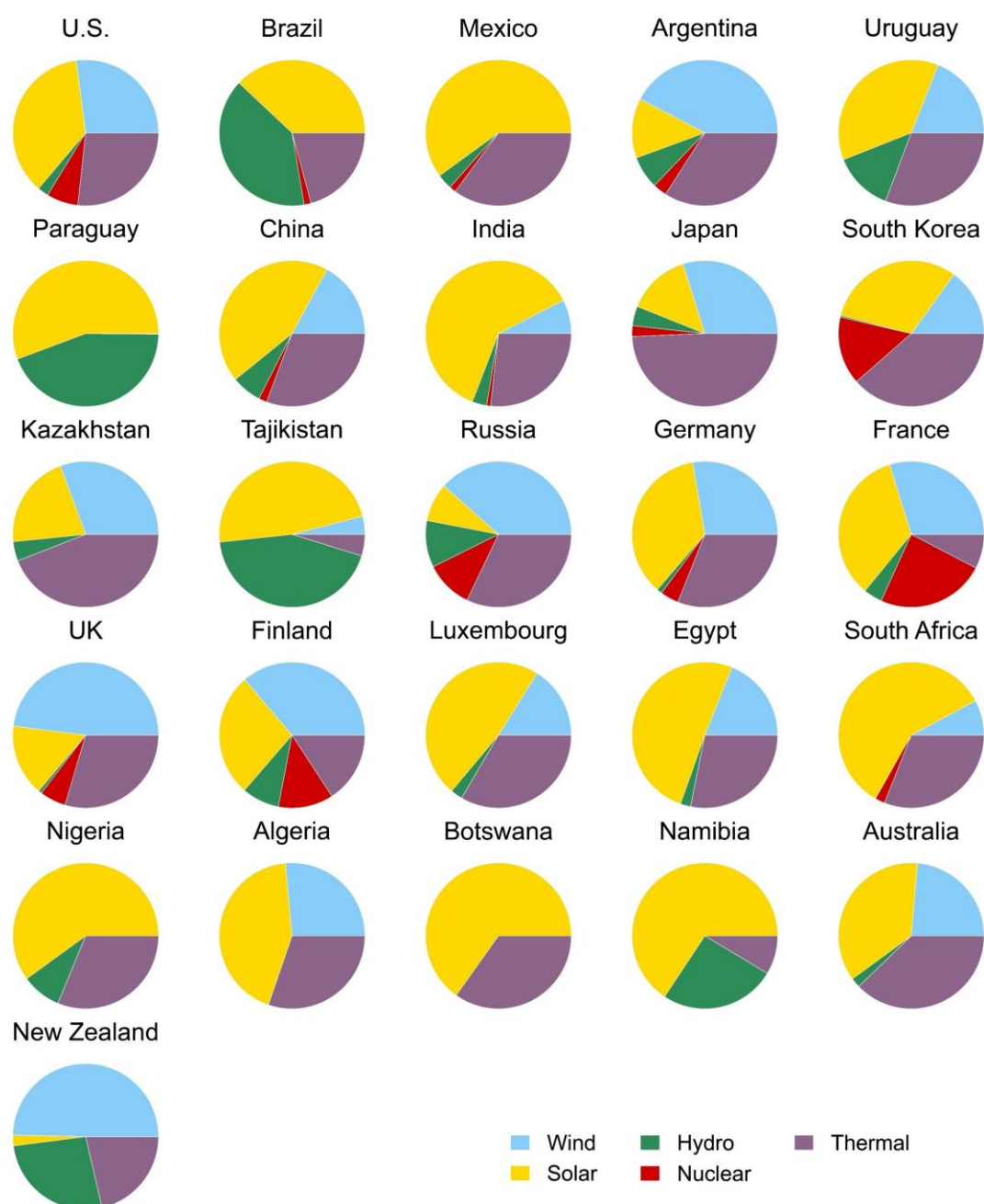
Supplementary Figure 4 | The seasonal and diurnal distributions of extreme hours across major countries under SSP126 scenario. In each case, the dots show the average values across dispatch model simulations with climate model ensembles; the vertical lines show the 33rd to 67th percentile ranges; the horizontal lines show the median values.



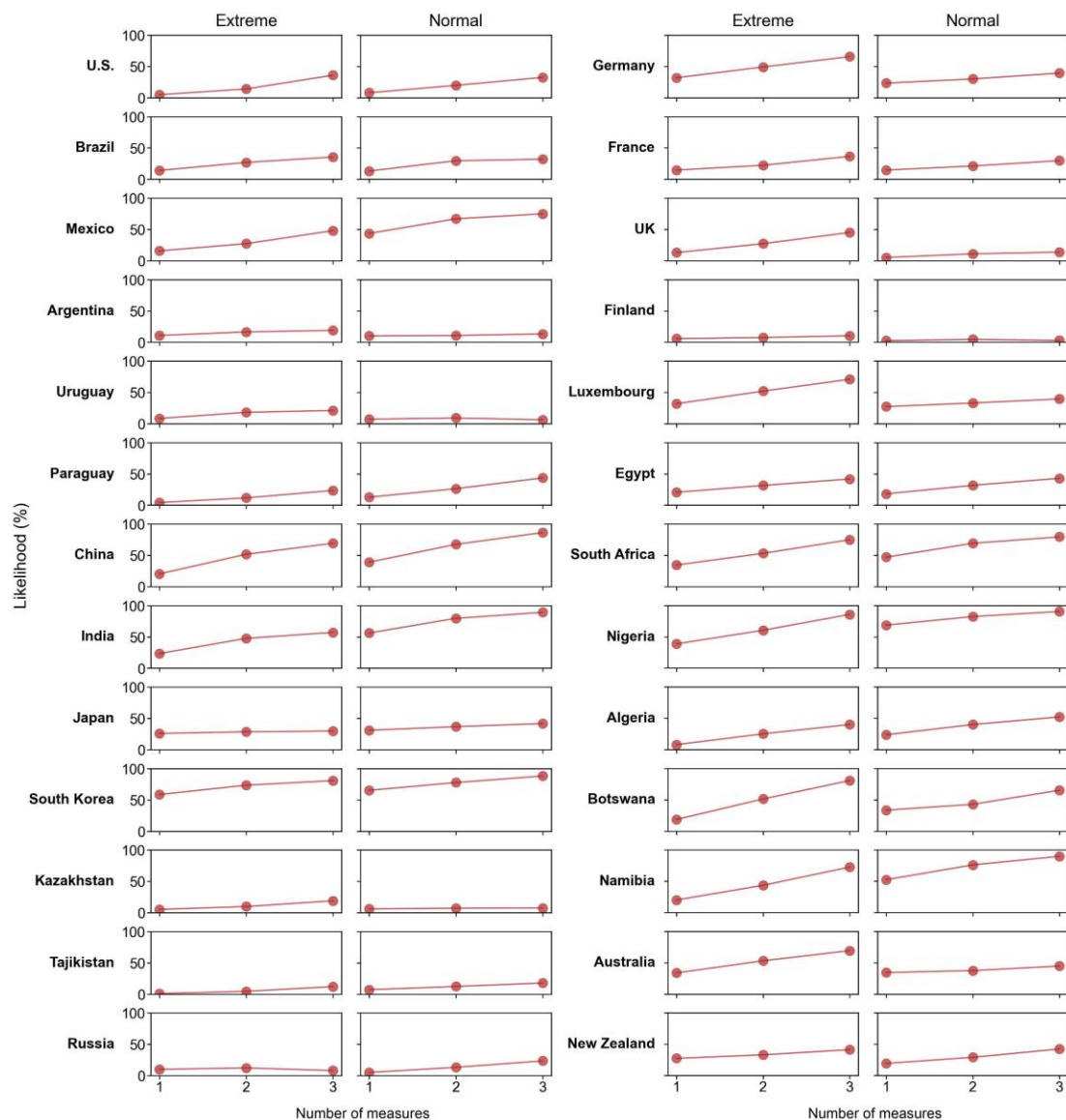
Supplementary Figure 5 | Median contributions to power hourly costs and supply under SSP245 scenario by different technologies. a–b, Median contributions to system costs (a) and median supply (b) across dispatch model simulations with climate model ensembles during extreme period. **c–d,** Median contributions to system costs (c) and median supply (d) during normal period. Striped and solid bars in (a) and (c) represent variable and fixed costs, respectively. It is noted that optimized wind/solar generation ratio varies across different years and climate models.



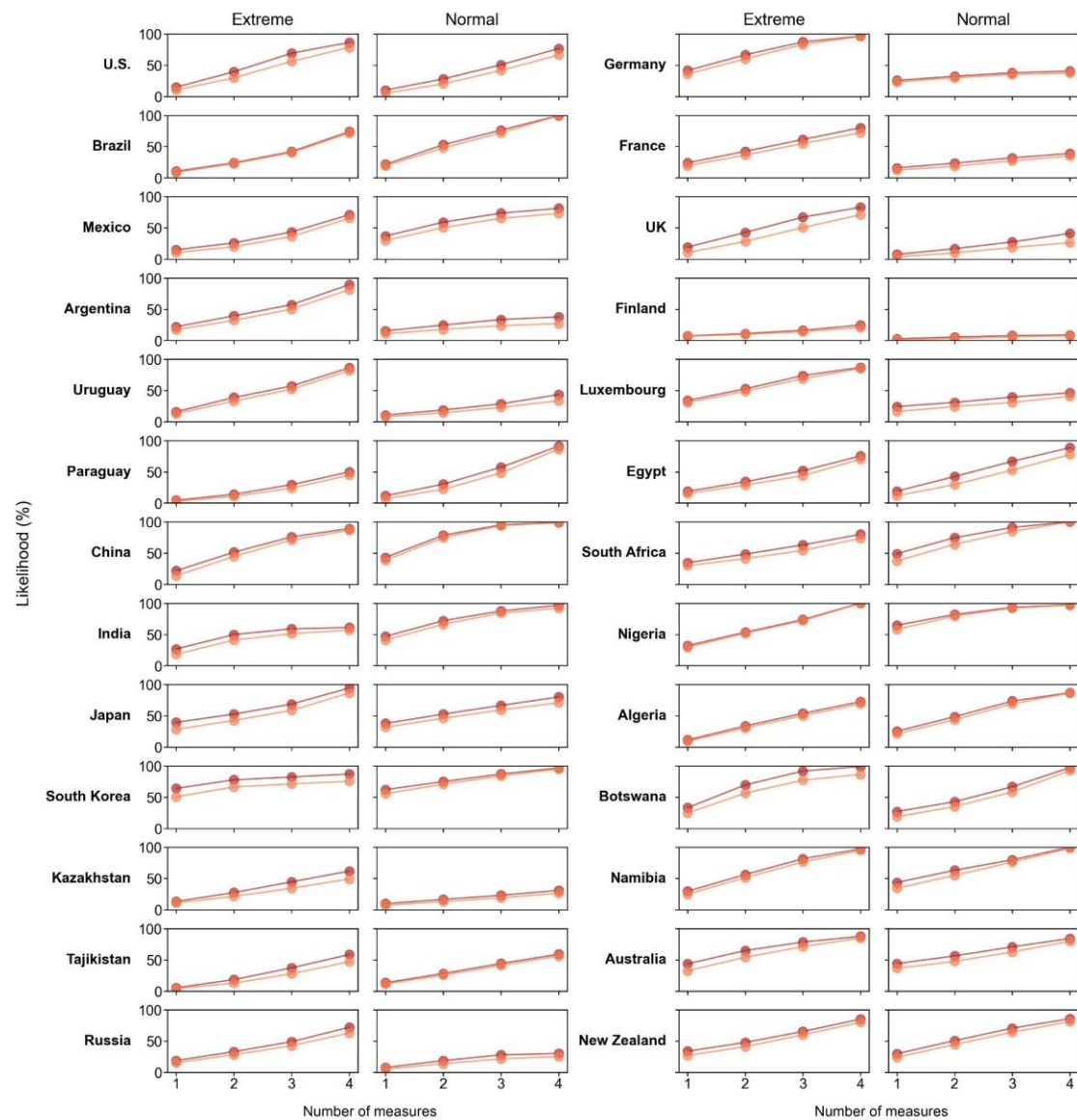
Supplementary Figure 6 | Changes in flexible energy capacity induced by climate change across 26 major countries under the SSP126 scenario. In each case, the bars show the average values across dispatch model simulations with climate model ensembles; the horizontal lines show the 33rd to 67th percentile ranges; the vertical lines show the median values. The changes in flexible energy capacity refers to the absolute difference between current climate (2016–2020) and future climate (2056–2060).



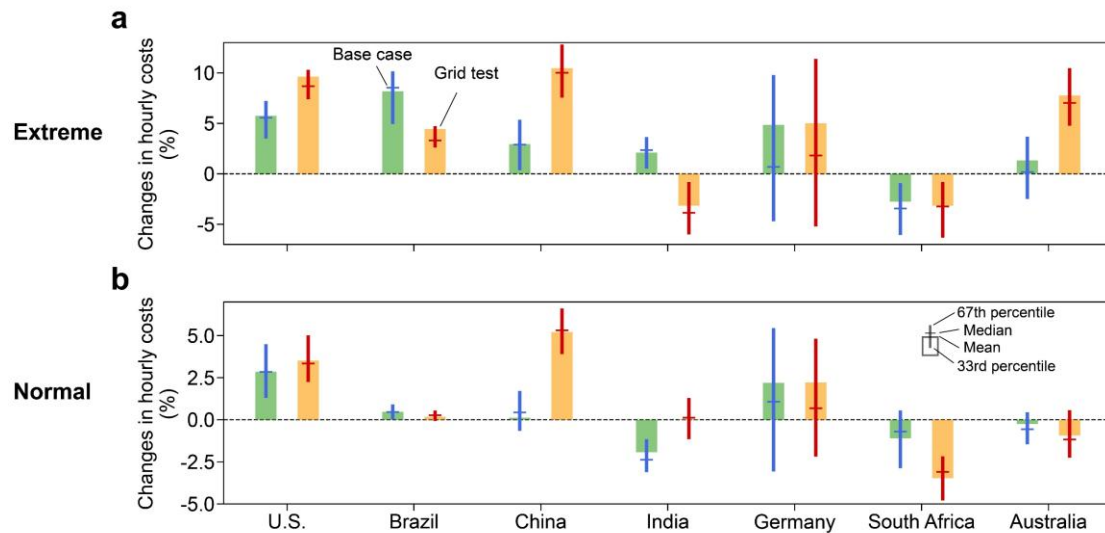
Supplementary Figure 7 | Median contributions to annual power generation for 26 major countries under the SSP126 scenario by different technologies across dispatch model simulations with climate model ensembles. It is noted that optimized wind/solar generation ratio varies across different years and climate models.



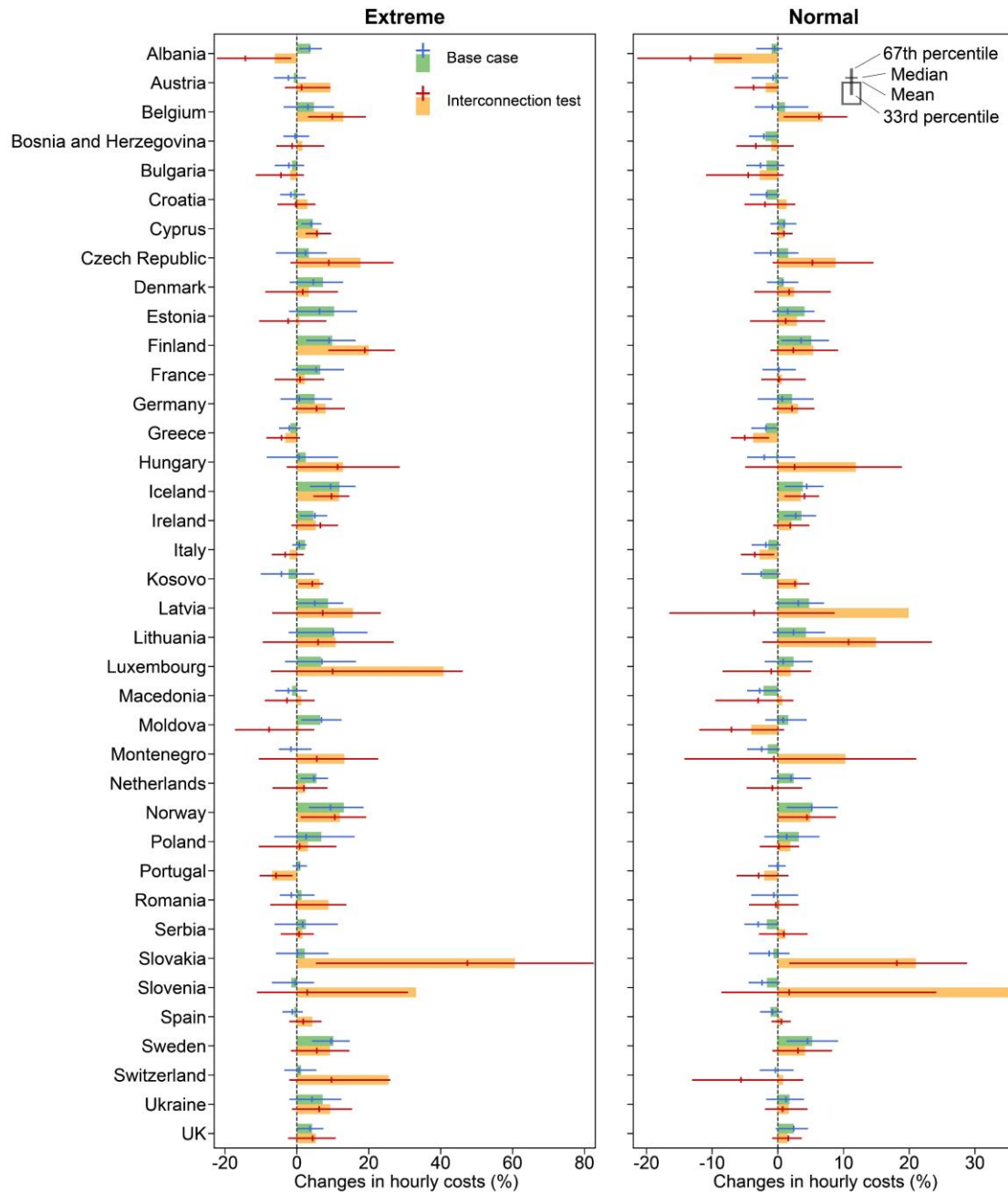
Supplementary Figure 8 | The likelihood of different numbers of measures mitigating the top 5% of events with the largest cost increases under SSP126 (excluding expanded interconnection measure). The red lines represent the relationship between the number of measures (ranging from 1 to 3) and the likelihood of mitigating hourly cost increases in the top 5% of events with the largest cost increases (i.e., those above the 95th percentile of dispatch model simulations with climate model ensembles) during extreme and normal periods.



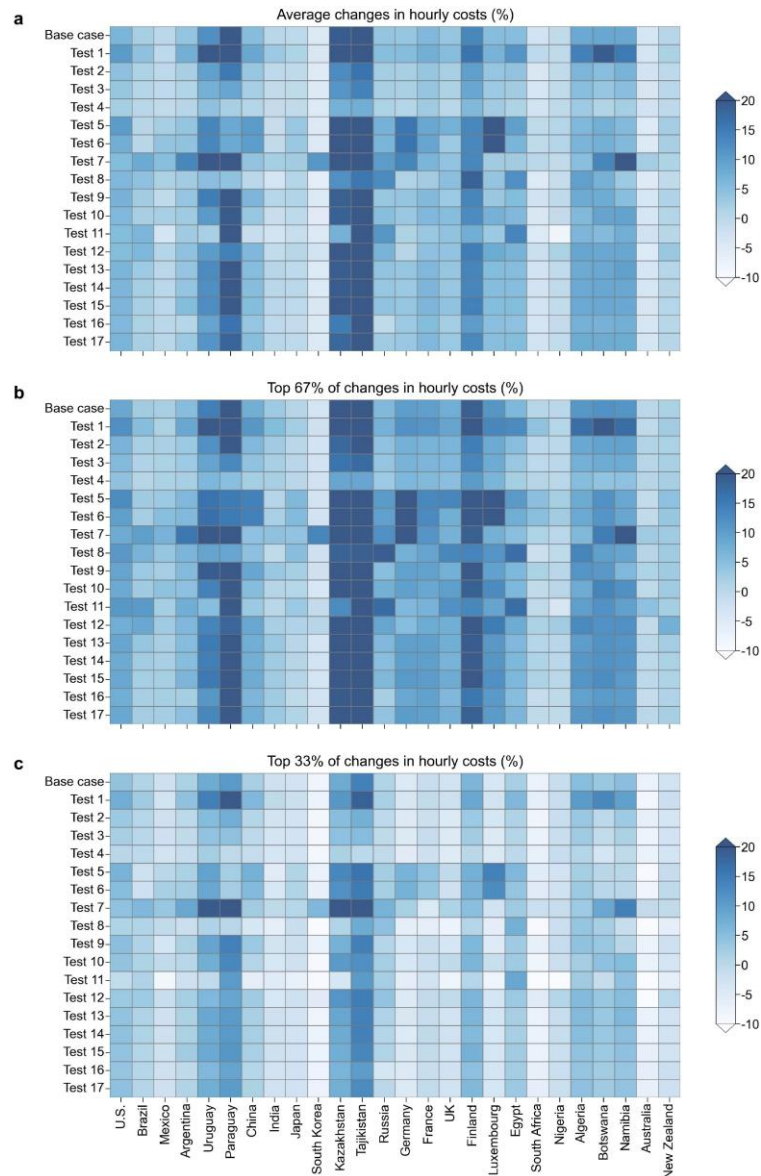
Supplementary Figure 9 | The likelihood of different numbers of measures addressing the top event and the top 5% of events with the largest cost increases for 26 major countries under SSP126. The line charts represent the relationship between the likelihood of mitigating hourly cost increases and the number of mitigation measures (ranging from 1 to 4). The red lines represent the top 5% of events with the largest cost increases (i.e., those above the 95th percentile of dispatch model simulations with climate model ensembles), while the orange lines represent the event with the largest cost increase.



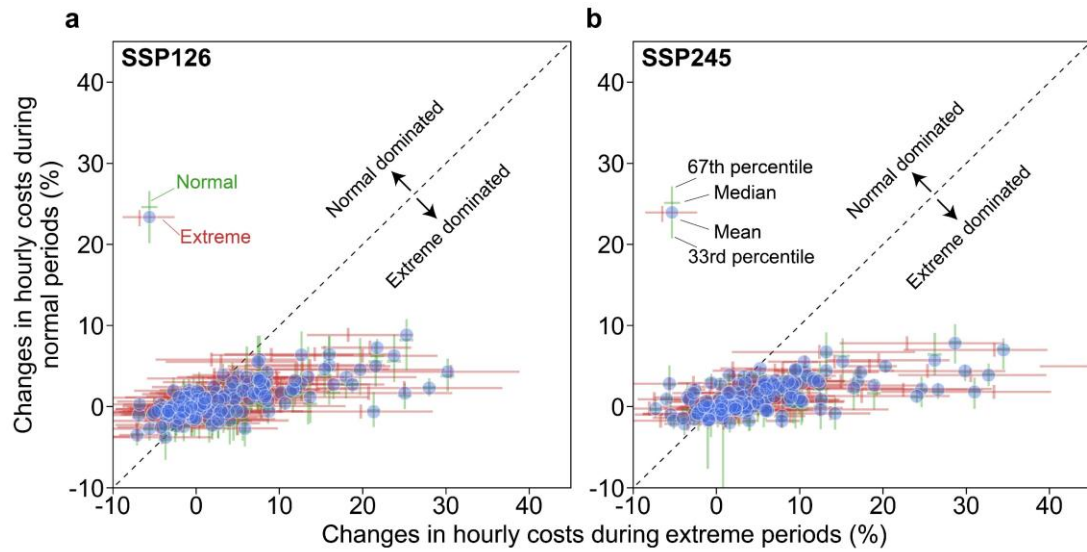
Supplementary Figure 10 | Changes in hourly costs under the SSP126 scenario by different technologies across U.S., Brazil, China, India, Germany, South Africa, and Australia, comparing the base case and the grid test. a–b, The changes in hourly costs during extreme **(a)** and normal periods **(b)**. In each case, the bars show the average values across dispatch model simulations with climate model ensembles; the vertical lines show the 33rd to 67th percentile ranges; the horizontal lines show the median values. The changes in hourly costs refer to the relative difference between current climate (2016–2020) and future climate (2056–2060).



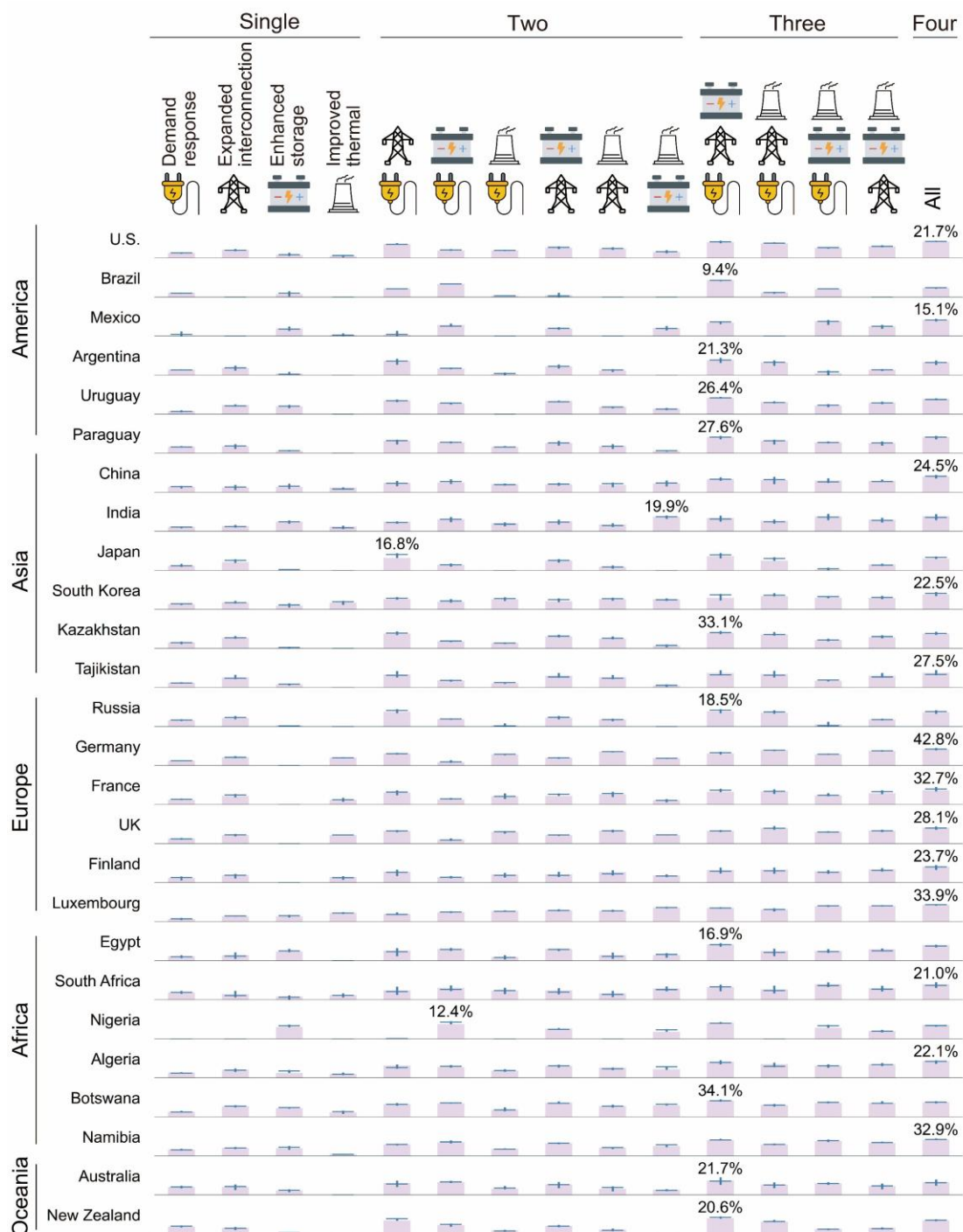
Supplementary Figure 11 | The comparison of the changes in hourly costs induced by climate change between the base case and the interconnection test across European countries under the SSP126 scenario. In each case, the bars show the average values across dispatch model simulations with climate model ensembles; the horizontal lines show the 33rd to 67th percentile ranges; the vertical lines show the median values. The changes in hourly costs refer to the relative difference between current climate (2016–2020) and future climate (2056–2060).



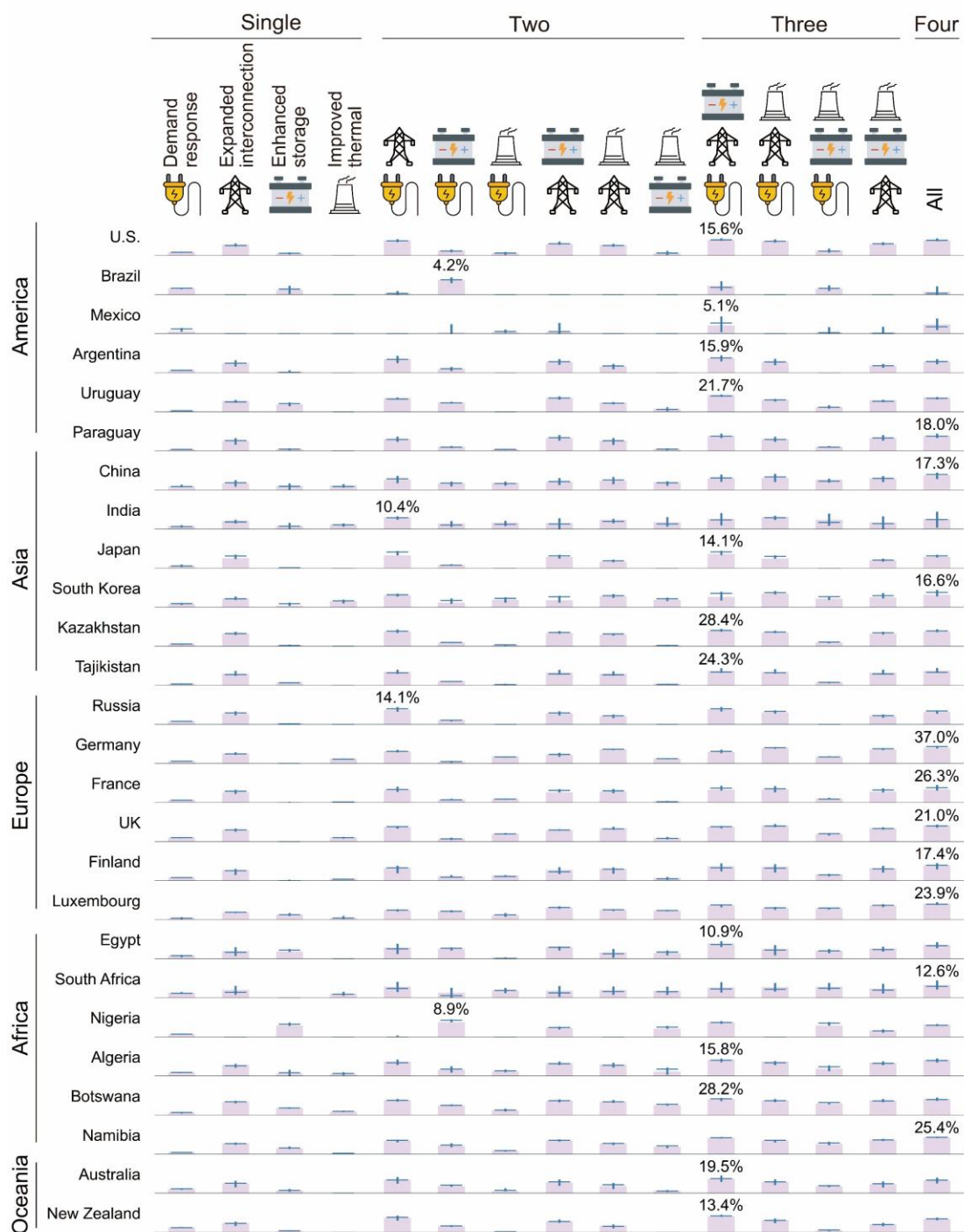
Supplementary Figure 12 | The comparison of the changes in hourly costs during extreme period induced by climate change among the base case and sensitivity tests under SSP126 scenario. The heatmaps show the average values **(a)**, 33rd percentiles **(b)**, 67th percentiles **(c)** of changes in hourly costs across dispatch model simulations with climate model ensembles. Test 1: aggressive cooling/heating coefficients; Test 2: conservative cooling/heating coefficients; Test 3: 70% of T-dependent demand share; Test 4: 50% of T-dependent demand share; Test 5: consideration of CCUS technology; Test 6: IAM-predicted wind/solar generation ratio; Test 7: 10th percentile of IAM-predicted wind and solar share; Test 8: 90th percentile of IAM-predicted wind and solar share; Test 9: a higher 10% energy storage cost; Test 10: a lower 10% energy storage cost; Test 11: 10th percentile of IAM-predicted thermal power share; Test 12: fixed hydropower profile as electricity demand; Test 13: a higher 10% hydro power; Test 14: nuclear power with the 80% minimum output limit; Test 15: fixed cost allocation method of *CF10th*; Test 16: fixed cost allocation method of *CFmax*; Test 17: fixed cost allocation method of *CFmax_weight*.



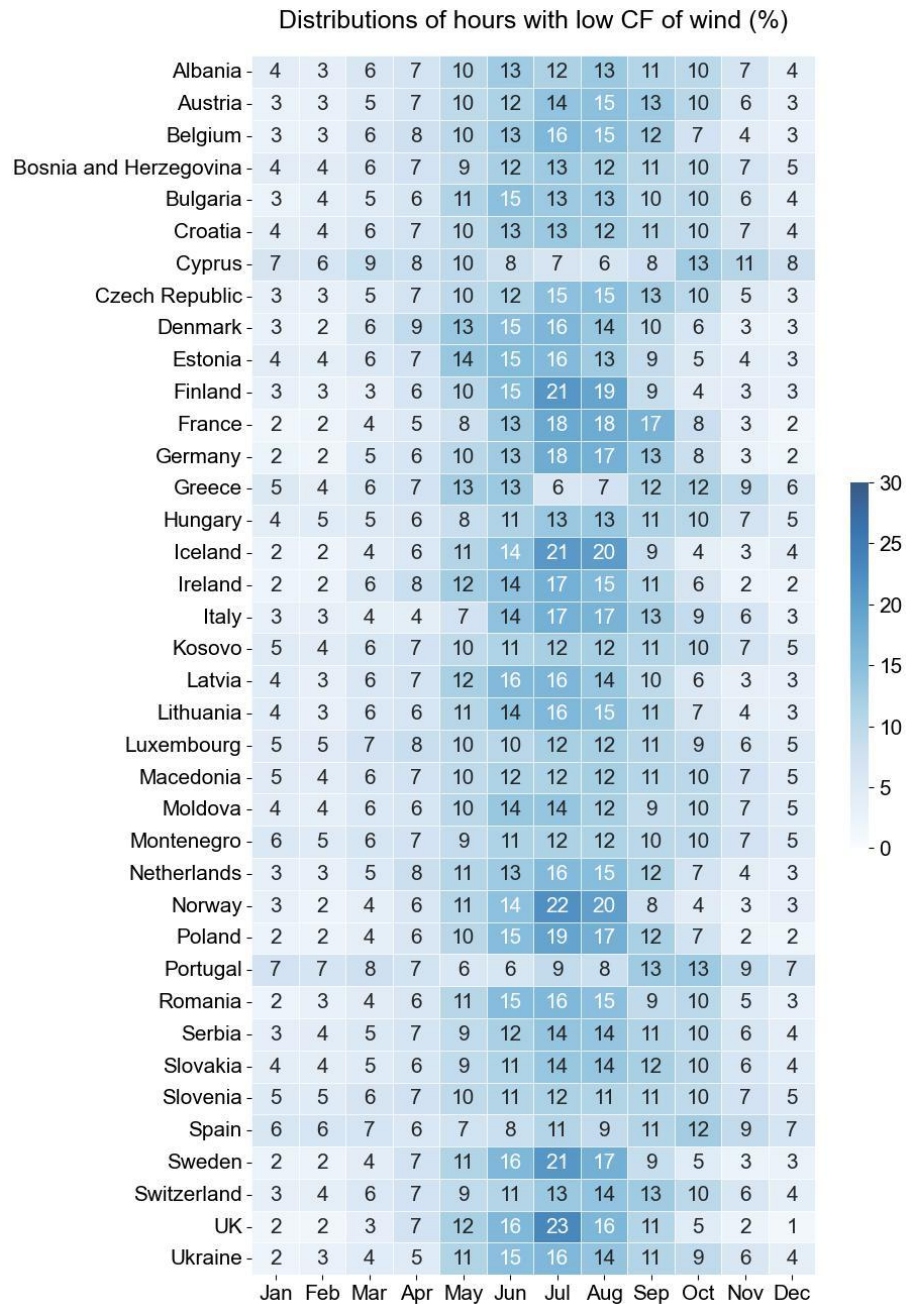
Supplementary Figure 13 | Changes in hourly costs under future climate change, with extreme periods defined as the 95th percentile of hourly costs. a–b, Changes in system costs during extreme and normal periods under the SSP126 (a) and SSP245 (b) scenarios for 178 countries. The scatterplots represent individual countries, showing average changes across dispatch model simulations with climate model ensembles. The horizontal red and vertical green lines show the 33rd to 67th percentile ranges during extreme and normal periods, respectively; the vertical red and horizontal green lines show the median changes during extreme and normal periods, respectively.



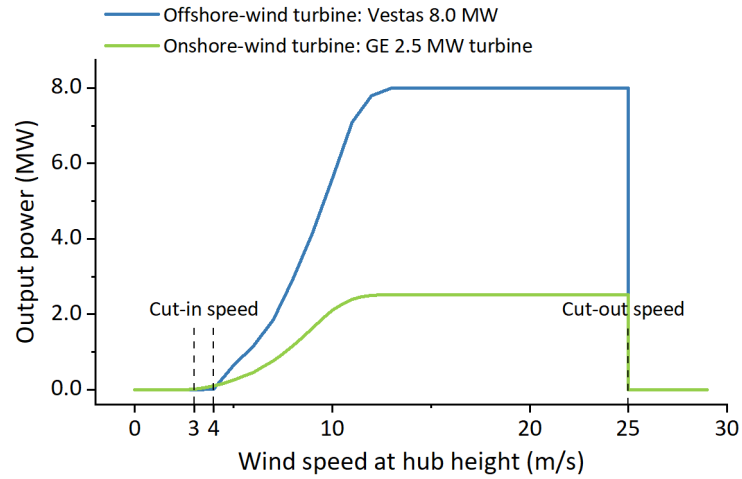
Supplementary Figure 14 | Reduction potential in future hourly costs for different combinations of measures during extreme periods under SSP126 scenario through aggressive strategies. In each case, the bars show the average cost reductions in hourly costs across dispatch model simulations with climate model ensembles; the vertical lines show the 33rd to 67th percentile ranges; the horizontal lines show the median values. The illustrations representing the four measures are available for free download in PNG format and were designed by Freepik (www.freepik.com).



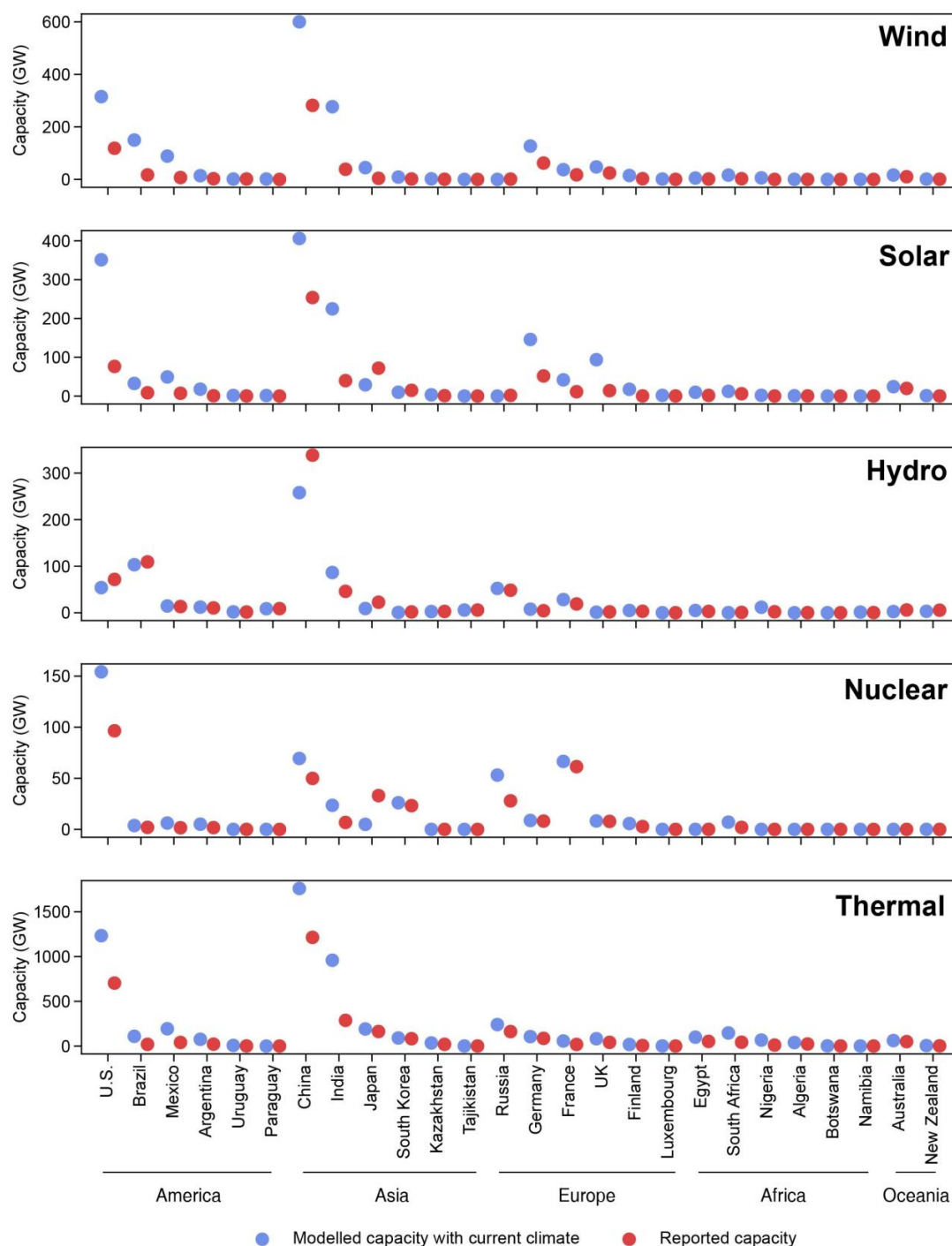
Supplementary Figure 15 | Reduction potential in future hourly costs for different combinations of measures during extreme periods under SSP126 scenario through conservative strategies. In each case, the bars show the average cost reductions in hourly costs across dispatch model simulations with climate model ensembles; the vertical lines show the 33rd to 67th percentile ranges; the horizontal lines show the median values. The illustrations representing the four measures are available for free download in PNG format and were designed by Freepik (www.freepik.com).



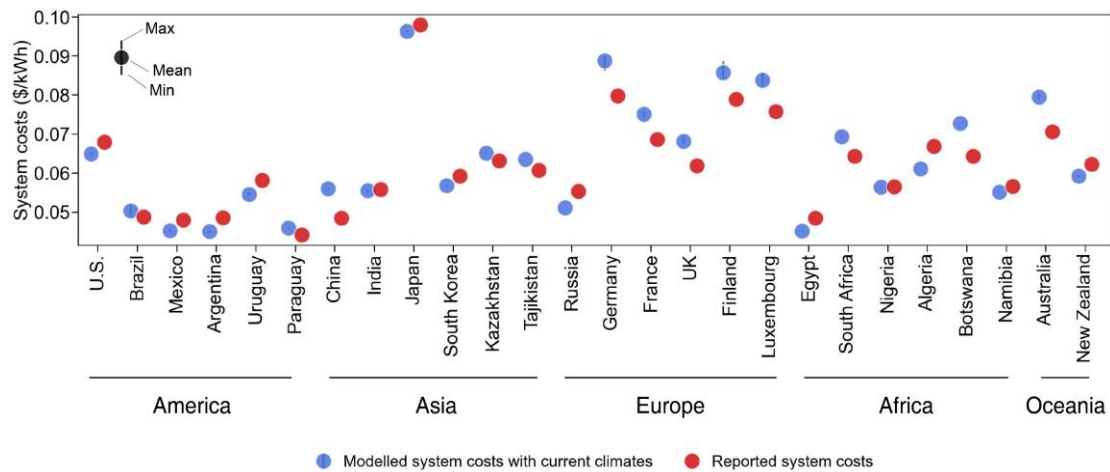
Supplementary Figure 16 | The monthly distributions of hours with low wind capacity factor (i.e., below 10th percentile of non-zero values) across European countries under SSP126 scenario. The heatmap shows the average values across multi-model ensembles.



Supplementary Figure 17 | The power curves of onshore and offshore wind turbines in this study. The specifications for onshore and offshore wind turbines are General Electric (GE) 2.5 MW and Vestas 8.0 MW, respectively. The cut-in speeds for onshore and offshore wind turbines are 3 m/s and 4 m/s, respectively, while the cut-out speed for both is 25 m/s, with all values at a hub height of 100 meters above ground.



Supplementary Figure 18 | The comparison of modelled installed capacity under the current climate and the reported installed capacity by different technologies. For each major country, the blue points represent the average capacities under current climate (2016–2020) driven by ERA5 reanalysis data, and the red points represent the reported installed capacities collected from IRENA (<https://pxweb.irena.org/pxweb/en/IRENASTAT/>).



Supplementary Figure 19 | Comparison of modelled system costs under the current climates, the reported system costs and future projections across major countries. For each major country, the blue points represent the average system costs under current climate (2016–2020) provided by ERA5, while the red points represent the reported system costs.

Supplementary Tables

Supplementary Table 1 | The surveyed 178 countries across the world

No.	Country	No.	Country	No.	Country	No.	Country
1	Afghanistan	46	Egypt	91	Liberia	136	Saudi Arabia
2	Albania	47	El Salvador	92	Libya	137	Senegal
3	Algeria	48	Equatorial Guinea	93	Lithuania	138	Serbia
4	Angola	49	Eritrea	94	Luxembourg	139	Seychelles
5	Antigua and Barbuda	50	Estonia	95	Macedonia	140	Sierra Leone
6	Argentina	51	Ethiopia	96	Madagascar	141	Singapore
7	Armenia	52	Fiji	97	Malawi	142	Slovakia
8	Australia	53	Finland	98	Malaysia	143	Slovenia
9	Austria	54	France	99	Mali	144	Solomon Islands
10	Azerbaijan	55	French Guiana	100	Malta	145	Somalia
11	Bahamas	56	Gabon	101	Mauritania	146	South Africa
12	Bahrain	57	Gambia	102	Mauritius	147	South Korea
13	Bangladesh	58	Georgia	103	Mexico	148	South Sudan
14	Belarus	59	Germany	104	Moldova	149	Spain
15	Belgium	60	Ghana	105	Mongolia	150	Sri Lanka
16	Belize	61	Greece	106	Montenegro	151	Sudan
17	Benin	62	Guatemala	107	Morocco	152	Suriname
18	Bhutan	63	Guinea	108	Mozambique	153	Swaziland
19	Bolivia	64	Guinea-Bissau	109	Myanmar	154	Sweden
20	Bosnia and Herzegovina	65	Guyana	110	Namibia	155	Switzerland
21	Botswana	66	Haiti	111	Nepal	156	Syria
22	Brazil	67	Honduras	112	Netherlands	157	Tajikistan
23	Brunei	68	Hungary	113	New Caledonia	158	Tanzania
24	Bulgaria	69	Iceland	114	New Zealand	159	Thailand
25	Burkina Faso	70	India	115	Nicaragua	160	Togo
26	Burundi	71	Indonesia	116	Niger	161	Trinidad and Tobago
27	Cambodia	72	Iran	117	Nigeria	162	Tunisia
28	Cameroon	73	Iraq	118	North Korea	163	Turkey
29	Canada	74	Ireland	119	Norway	164	Turkmenistan
30	Cape Verde	75	Israel	120	Oman	165	Uganda
31	Central African Republic	76	Italy	121	Pakistan	166	Ukraine
32	Chad	77	Ivory Coast	122	Panama	167	United Arab Emirates

No.	Country	No.	Country	No.	Country	No.	Country
33	Chile	78	Jamaica	123	Papua New Guinea	168	UK
34	China	79	Japan	124	Paraguay	169	U.S.
35	Colombia	80	Jordan	125	Peru	170	Uruguay
36	Costa Rica	81	Kazakhstan	126	Philippines	171	Uzbekistan
37	Croatia	82	Kenya	127	Poland	172	Vanuatu
38	Cuba	83	Kiribati	128	Portugal	173	Venezuela
39	Cyprus	84	Kosovo	129	Puerto Rico	174	Vietnam
40	Czech Republic	85	Kuwait	130	Qatar	175	Western Sahara
41	Democratic Republic of the Congo	86	Kyrgyzstan	131	Republic of Congo	176	Yemen
42	Denmark	87	Laos	132	Romania	177	Zambia
43	Djibouti	88	Latvia	133	Russia	178	Zimbabwe
44	Dominican Republic	89	Lebanon	134	Rwanda		
45	Ecuador	90	Lesotho	135	Sao Tome and Principe		

Supplementary Table 2 | Summary of model simulations

Name	Climate scenario	Country	Period	Hour	Case number
SSP126-base	SSP126 (7 models)	178	2016-2020 2056-2060	8,760h	12,460 (178 countries×10 years×7 models)
SSP245-base	SSP245 (7 models)	178	2016-2020 2056-2060	8,760h	12,460 (178×10×7)
SSP126-solu	SSP126 (15 strategies)	26	2016-2020 2056-2060	8,760h	27,300 (26×10×7×15)
SSP245-solu	SSP245 (15 strategies)	26	2016-2020 2056-2060	8,760h	27,300 (26×10×7×15)
Nested with transmission					
SSP126-nested	SSP126 (With transmission)	7	2016-2020 2056-2060	2,920h (3h interval)	490 (7×10×7)
SSP126-comp	SSP126 (Without transmission)	7	2016-2020 2056-2060	2,920h	490 (7×10×7)
SSP126-nested-EU	SSP126 (With transmission)	38	2016-2020 2056-2060	2,920h	70 (1×10×7)
SSP126-comp-EU	SSP126 (Without transmission)	38	2016-2020 2056-2060	2,920h	2,660 (38×10×7)
Sensitivity tests					
Sen-current cost	ERA5	26	2016-2020	8,760h	130 (26×5)
Sen-wind-solar supply share	SSP126 (90th+10th)	26	2016-2020 2056-2060	8,760h	3,640 (26×10×7×2)
Sen-wind/solar-ratio	SSP126	26	2016-2020 2056-2060	8,760h	1,820 (26×10×7)
Sen-storage	SSP126 (Low + high)	26	2016-2020 2056-2060	8,760h	3,640 (26×10×7×2)
Sen-low thermal	SSP126	26	2016-2020 2056-2060	8,760h	1,820 (26×10×7)
Sen-nuclear flexibility	SSP126	26	2016-2020 2056-2060	8,760h	1,820 (26×10×7)
Sen-hydro increase	SSP126	26	2016-2020 2056-2060	8,760h	1,820 (26×10×7)
Sen-add CCUS	SSP126	26	2016-2020 2056-2060	8,760h	1,820 (26×10×7)
Sen-fixed hydro profile	SSP126	26	2016-2020 2056-2060	8,760h	1,820 (26×10×7)
Sen-temperature	SSP126 (Low + high)	26	2016-2020 2056-2060	8,760h	3,640 (26×10×7×2)
Sen-T-dependent demand share	SSP126 (50% + 70%)	26	2016-2020 2056-2060	8,760h	3,640 (26×10×7×2)
Sen-solutions	SSP126 (Aggressive + conservative)	26	2016-2020 2056-2060	8,760h	54,600 (26×10×7×15×2)
Total					163,440

Supplementary Table 3 | The selected 26 countries

Continent	Country
America	U.S., Brazil, Mexico, Argentina, Paraguay, Uruguay
Asia	China, India, Japan, South Korea, Kyrgyzstan, Tajikistan
Europe	Russia, Germany, France, UK, Finland, Luxembourg
Africa	Egypt, South Africa, Nigeria, Algeria, Namibia, Botswana
Oceania	Australia, New Zealand

Supplementary Table 4 | The average annual total additional climate-related system cost and GDP loss across dispatch model simulations with climate model ensembles under SSP126 and SSP245 scenarios

Region	Country	Changes in annual system cost (billion \$)		GDP loss (%)	
		SSP126	SSP245	SSP126	SSP245
America	U.S.	15.75	10.33	0.04	0.03
	Brazil	3.51	4.59	0.10	0.14
	Mexico	1.86	2.40	0.06	0.08
	Argentina	0.91	0.98	0.09	0.10
	Uruguay	0.21	0.21	0.15	0.14
	Paraguay	0.33	0.32	0.35	0.33
Asia	China	5.11	12.43	0.01	0.03
	India	10.10	22.50	0.05	0.12
	Japan	-0.01	1.68	0.00	0.02
	South Korea	-2.05	-0.74	-0.06	-0.02
	Kazakhstan	1.48	0.79	0.42	0.22
	Tajikistan	0.26	0.19	1.57	1.13
Europe	Russia	1.63	-2.26	0.05	-0.07
	Germany	1.34	0.17	0.02	0.00
	France	1.44	0.91	0.03	0.02
	UK	0.91	0.40	0.02	0.01
	Finland	1.07	0.26	0.22	0.05
	Luxembourg	0.02	-0.01	0.01	0.00
Africa	Egypt	1.71	1.97	0.15	0.18
	South Africa	-0.08	0.90	-0.01	0.09
	Nigeria	0.59	1.29	0.05	0.10
	Algeria	0.87	0.96	0.18	0.20
	Botswana	0.05	0.08	0.12	0.19
	Namibia	0.04	0.07	0.12	0.22
Oceania	Australia	0.47	0.36	0.01	0.01
	New Zealand	0.00	-0.01	0.00	0.00

Supplementary Table 5 | The 95th percentiles of annual total additional climate-related system cost and GDP loss across dispatch model simulations with climate model ensembles under SSP126 and SSP245 scenarios

Region	Country	Changes in annual system cost (billion \$)		GDP loss (%)	
		SSP126	SSP245	SSP126	SSP245
America	U.S.	38.05	23.76	0.09	0.06
	Brazil	9.23	9.44	0.27	0.28
	Mexico	5.51	5.05	0.18	0.16
	Argentina	2.49	2.20	0.25	0.22
	Uruguay	0.42	0.34	0.29	0.23
	Paraguay	0.64	0.56	0.67	0.59
Asia	China	36.28	41.05	0.08	0.09
	India	31.57	38.58	0.16	0.20
	Japan	4.97	4.65	0.07	0.07
	South Korea	1.14	0.67	0.03	0.02
	Kazakhstan	4.19	1.82	1.19	0.52
	Tajikistan	0.61	0.34	3.65	2.00
Europe	Russia	17.29	1.63	0.56	0.05
	Germany	17.45	4.45	0.24	0.06
	France	10.95	6.20	0.23	0.13
	UK	5.54	2.06	0.11	0.04
	Finland	4.24	1.71	0.86	0.34
	Luxembourg	0.12	0.03	0.09	0.02
Africa	Egypt	3.86	2.89	0.35	0.26
	South Africa	3.51	3.98	0.36	0.40
	Nigeria	1.34	2.08	0.11	0.17
	Algeria	1.46	1.34	0.30	0.28
	Botswana	0.11	0.15	0.26	0.34
	Namibia	0.08	0.10	0.25	0.34
Oceania	Australia	2.64	1.34	0.07	0.03
	New Zealand	0.15	0.19	0.02	0.03

Supplementary Table 6 | Sources and basics of climate variables

Scenario	Variable	Global climate model	Resolution (Latitude × longitude)	Frequency	Period	Institution	Link	Last access	Reference
SSP126 SSP245	Near-surface air temperature (<i>tas</i>), downwelling shortwave irradiance (<i>rsds</i>), surface downwelling shortwave flux in air assuming clear sky (<i>rsdscs</i>), eastward wind (<i>uas</i>), and northward wind (<i>vas</i>)	BCC-CSM2-MR	1.125°×1.125°	3hr	2016–2020 and 2056–2060	China Meteorological Administration (China)	https://www.wdc-climate.de/ui/entry?acronym=C6SPBCBCM	2019-03-08	Xin et al., 2023 ³¹
		CanESM5	2.8125°×2.8125°			Canadian Centre for Climate Modelling and Analysis (Canada)	https://www.wdc-climate.de/ui/cmip6?input=CMIP6.ScenarioMIP.CCCma.CanESM5	2019-03-06	Swart, et al., 2019 ³²
		MIROC6	1.410°×1.410°			Japanese Modeling Community (Japan)	https://www.wdc-climate.de/ui/cmip6?input=CMIP6.ScenarioMIP.MIROC.MIROC6	2019-06-27	Shiogama et al., 2019 ³³
		MIROC-ES2H	1.410°×1.410°			Japanese Modeling Community (Japan)	https://www.wdc-climate.de/ui/entry?acronym=C6CMMI MIH	2021-01-25	Watanabe et al., 2021 ³⁴
		MPI-ESM1-2-HR	0.9375°×0.9375°			Max Planck Institute for Meteorology (Germany)	https://www.wdc-climate.de/ui/cmip6?input=CMIP6.HighResMIP.MPI-M.MPI-ESM1-2-HR	2019-07-10	von Storch et al., 2017 ³⁵
		MRI-ESM2-0	1.125°×1.125°			Meteorological Research Institute (Japan)	https://www.wdc-climate.de/ui/cmip6?input=CMIP6.CMIP.MRI.MRI-ESM2-0	2019-02-22	Yukimoto et al., 2023 ³⁶
		NESM3	1.875°×1.875°			Nanjing University of Information Science & Technology (China)	https://www.wdc-climate.de/ui/entry?acronym=C6SPNUI NES	2019-07-24	Cao, 2023 ³⁷

Supplementary Table 7 | Data sources of land cover, elevation, slope, and natural reserves

Type	Data source	Link	Reference
Land cover	The European Space Agency's Climate Change Initiative Land cover dataset	http://maps.elie.ucl.ac.be/CCI/viewer/download.php	ESA, 2017 ³⁸
Elevation and slope	The Global Multi-resolution Terrain Elevation Data Global Grids mn30	https://topotools.cr.usgs.gov/gmted_viewer/gmted2010_global_grids.php	Danielson & Gesch, 2011 ³⁹
Natural reserves	The World Database on Protected Areas	https://www.protectedplanet.net	UNEP-WCMC & IUCN, 2025 ⁴⁰

Supplementary Table 8 | Specifications for assumed onshore and offshore wind turbines

Specification	GE 2.5 MW	Vestas 8.0 MW
Rated power	2,500.0 kW	8,000.0 kW
Cut-in wind speed	3.0 m/s	4.0 m/s
Rated power speed	13.0 m/s	13.0 m/s
Cut-out wind speed	25.0 m/s	25.0 m/s
Diameter	100.0 m	164.0 m
Swept area	7,853.9 m ²	21,164.1 m ²
Number of blades	3	3
Hub height	100.0 m	100.0 m

Supplementary Table 9 | IAM-predicted wind and solar generation share by 2060 under the SSP126 and SSP245 scenarios

Country	SSP126	SSP245	Region
Afghanistan	71.20%	52.70%	R10INDIA+
Albania	63.80%	55.50%	R10EUROPE
Algeria	70.50%	41.80%	R10MIDDLE_EAST
Angola	65.40%	59.20%	R10AFRICA
Antigua and Barbuda	56.10%	47.10%	R10NORTH_AM
Argentina	56.10%	47.10%	R10LATIN_AM
Armenia	50.90%	33.00%	R10REF_ECON
Australia	59.80%	55.60%	R10PAC_OECD
Austria	63.80%	55.50%	R10EUROPE
Azerbaijan	60.40%	45.50%	R10REST_ASIA
Bahamas	56.10%	47.10%	R10LATIN_AM
Bahrain	70.50%	41.80%	R10MIDDLE_EAST
Bangladesh	71.20%	52.70%	R10INDIA+
Belarus	63.80%	55.50%	R10EUROPE
Belgium	63.80%	55.50%	R10EUROPE
Belize	56.10%	47.10%	R10LATIN_AM
Benin	65.70%	50.30%	R10AFRICA
Bhutan	71.20%	52.70%	R10INDIA+
Bolivia	56.10%	47.10%	R10LATIN_AM
Bosnia and Herzegovina	63.80%	55.50%	R10EUROPE
Botswana	65.70%	50.30%	R10AFRICA
Brazil	38.00%	36.60%	R10LATIN_AM
Brunei	60.40%	45.50%	R10REST_ASIA
Bulgaria	63.80%	55.50%	R10EUROPE
Burkina Faso	65.70%	50.30%	R10AFRICA
Burundi	65.70%	50.30%	R10AFRICA
Cambodia	62.80%	50.00%	R10CHINA+
Cameroon	65.70%	50.30%	R10AFRICA
Canada	45.00%	41.00%	R10NORTH_AM
Cape Verde	65.70%	50.30%	R10AFRICA
Central African Republic	65.70%	50.30%	R10AFRICA
Chad	65.70%	50.30%	R10AFRICA
Chile	52.40%	47.10%	R10LATIN_AM
China	60.80%	48.10%	R10CHINA+
Colombia	52.20%	47.10%	R10LATIN_AM
Costa Rica	56.10%	47.10%	R10LATIN_AM
Croatia	63.80%	55.50%	R10EUROPE
Cuba	56.10%	47.10%	R10LATIN_AM
Cyprus	63.80%	55.50%	R10EUROPE
Czech Republic	63.80%	55.50%	R10EUROPE
Democratic Republic of the Congo	65.70%	50.30%	R10AFRICA
Denmark	63.80%	55.50%	R10EUROPE
Djibouti	65.70%	50.30%	R10AFRICA
Dominican Republic	56.10%	47.10%	R10LATIN_AM
Ecuador	56.10%	47.10%	R10LATIN_AM

Country	SSP126	SSP245	Region
Egypt	70.50%	41.80%	R10MIDDLE_EAST
El Salvador	56.10%	47.10%	R10LATIN_AM
Equatorial Guinea	65.70%	50.30%	R10AFRICA
Eritrea	65.70%	50.30%	R10AFRICA
Estonia	63.80%	55.50%	R10EUROPE
Ethiopia	60.90%	50.30%	R10AFRICA
Fiji	60.40%	45.50%	R10REST_ASIA
Finland	63.80%	55.50%	R10EUROPE
France	63.80%	55.50%	R10EUROPE
French Guiana	56.10%	47.10%	R10LATIN_AM
Gabon	65.70%	50.30%	R10AFRICA
Gambia	65.70%	50.30%	R10AFRICA
Georgia	63.80%	55.50%	R10EUROPE
Germany	63.80%	55.50%	R10EUROPE
Ghana	65.70%	50.30%	R10AFRICA
Greece	63.80%	55.50%	R10EUROPE
Guatemala	56.10%	47.10%	R10LATIN_AM
Guinea	65.70%	50.30%	R10AFRICA
Guinea-Bissau	65.70%	50.30%	R10AFRICA
Guyana	56.10%	47.10%	R10LATIN_AM
Haiti	56.10%	47.10%	R10LATIN_AM
Honduras	56.10%	47.10%	R10LATIN_AM
Hungary	63.80%	55.50%	R10EUROPE
Iceland	63.80%	55.50%	R10EUROPE
India	69.40%	51.30%	R10INDIA+
Indonesia	50.90%	41.80%	R10REST_ASIA
Iran	70.50%	41.80%	R10MIDDLE_EAST
Iraq	70.50%	41.80%	R10MIDDLE_EAST
Ireland	63.80%	55.50%	R10EUROPE
Israel	70.50%	41.80%	R10MIDDLE_EAST
Italy	63.80%	55.50%	R10EUROPE
Ivory Coast	65.70%	50.30%	R10AFRICA
Jamaica	56.10%	47.10%	R10LATIN_AM
Japan	44.10%	33.80%	R10PAC_OECD
Jordan	63.80%	55.50%	R10EUROPE
Kazakhstan	50.90%	33.00%	R10REF_ECON
Kenya	43.20%	28.90%	R10AFRICA
Kiribati	51.60%	42.30%	R10PAC_OECD
Kosovo	63.80%	55.50%	R10EUROPE
Kuwait	70.50%	41.80%	R10MIDDLE_EAST
Kyrgyzstan	50.90%	33.00%	R10REF_ECON
Laos	62.80%	50.00%	R10CHINA+
Latvia	63.80%	55.50%	R10EUROPE
Lebanon	70.50%	41.80%	R10MIDDLE_EAST
Lesotho	65.70%	50.30%	R10AFRICA
Liberia	70.50%	41.80%	R10MIDDLE_EAST
Libya	96.60%	95.40%	R10MIDDLE_EAST
Lithuania	63.80%	55.50%	R10EUROPE

Country	SSP126	SSP245	Region
Luxembourg	63.80%	55.50%	R10EUROPE
Macedonia	63.80%	55.50%	R10EUROPE
Madagascar	66.80%	50.30%	R10AFRICA
Malawi	65.70%	50.30%	R10AFRICA
Malaysia	60.40%	45.50%	R10REST_ASIA
Mali	65.70%	50.30%	R10AFRICA
Malta	63.80%	55.50%	R10EUROPE
Mauritania	65.70%	50.30%	R10AFRICA
Mauritius	65.70%	50.30%	R10AFRICA
Mexico	59.70%	53.00%	R10LATIN_AM
Moldova	63.80%	55.50%	R10EUROPE
Mongolia	62.80%	50.00%	R10CHINA+
Montenegro	63.80%	55.50%	R10EUROPE
Morocco	76.20%	73.80%	R10MIDDLE_EAST
Mozambique	65.70%	50.30%	R10AFRICA
Myanmar	60.40%	45.50%	R10REST_ASIA
Namibia	65.70%	50.30%	R10AFRICA
Nepal	71.20%	52.70%	R10INDIA+
Netherlands	63.80%	55.50%	R10EUROPE
New Caledonia	51.60%	42.30%	R10PAC_OECD
New Zealand	51.60%	42.30%	R10PAC_OECD
Nicaragua	56.10%	47.10%	R10LATIN_AM
Niger	65.70%	50.30%	R10AFRICA
Nigeria	59.90%	50.20%	R10AFRICA
North Korea	62.80%	50.00%	R10CHINA+
Norway	63.80%	55.50%	R10EUROPE
Oman	70.50%	41.80%	R10MIDDLE_EAST
Pakistan	60.10%	52.70%	R10INDIA+
Panama	56.10%	47.10%	R10LATIN_AM
Papua New Guinea	51.60%	42.30%	R10PAC_OECD
Paraguay	56.10%	47.10%	R10LATIN_AM
Peru	56.10%	47.10%	R10LATIN_AM
Philippines	60.40%	45.50%	R10REST_ASIA
Poland	63.80%	55.50%	R10EUROPE
Portugal	63.80%	55.50%	R10EUROPE
Puerto Rico	56.10%	47.10%	R10LATIN_AM
Qatar	70.50%	41.80%	R10MIDDLE_EAST
Republic of Congo	65.70%	50.30%	R10AFRICA
Romania	63.80%	55.50%	R10EUROPE
Russia	47.20%	29.30%	R10REF_ECON
Rwanda	65.70%	50.30%	R10AFRICA
Sao Tome and Principe	65.70%	50.30%	R10AFRICA
Saudi Arabia	46.00%	17.90%	R10MIDDLE_EAST
Senegal	65.70%	50.30%	R10AFRICA
Serbia	63.80%	55.50%	R10EUROPE
Seychelles	65.70%	50.30%	R10AFRICA
Sierra Leone	65.70%	50.3%	R10AFRICA
Singapore	60.40%	45.5%	R10REST_ASIA

Country	SSP126	SSP245	Region
Slovakia	63.80%	55.5%	R10EUROPE
Slovenia	63.80%	55.5%	R10EUROPE
Solomon Islands	51.60%	42.3%	R10PAC_OECD
Somalia	65.70%	50.3%	R10AFRICA
South Africa	67.10%	51.5%	R10AFRICA
South Korea	46.10%	39.70%	R10PAC_OECD
South Sudan	65.70%	50.30%	R10AFRICA
Spain	63.80%	55.50%	R10EUROPE
Sri Lanka	71.20%	52.70%	R10INDIA+
Sudan	65.70%	50.30%	R10AFRICA
Suriname	56.10%	47.10%	R10LATIN_AM
Swaziland	65.70%	50.30%	R10AFRICA
Sweden	63.80%	55.50%	R10EUROPE
Switzerland	63.80%	55.50%	R10EUROPE
Syria	70.50%	41.80%	R10MIDDLE_EAST
Tajikistan	50.90%	33.00%	R10REF_ECON
Tanzania	65.70%	50.30%	R10AFRICA
Thailand	60.40%	45.50%	R10REST_ASIA
Togo	65.70%	50.30%	R10AFRICA
Trinidad and Tobago	56.10%	47.10%	R10LATIN_AM
Tunisia	70.50%	41.80%	R10MIDDLE_EAST
Turkey	52.40%	40.20%	R10EUROPE
Turkmenistan	50.90%	33.00%	R10REF_ECON
Uganda	65.70%	50.30%	R10AFRICA
Ukraine	63.80%	55.50%	R10EUROPE
United Arab Emirates	70.50%	41.80%	R10MIDDLE_EAST
UK	63.80%	55.50%	R10EUROPE
U.S.	64.00%	56.20%	R10LATIN_AM
Uruguay	56.10%	47.10%	R10LATIN_AM
Uzbekistan	50.90%	33.00%	R10REF_ECON
Vanuatu	51.60%	42.30%	R10PAC_OECD
Venezuela	47.50%	44.80%	R10LATIN_AM
Vietnam	62.80%	50.00%	R10CHINA+
Western Sahara	65.70%	50.30%	R10AFRICA
Yemen	70.50%	41.80%	R10MIDDLE_EAST
Zambia	65.70%	50.30%	R10AFRICA
Zimbabwe	65.70%	50.30%	R10AFRICA

Supplementary Table 10 | Data sources of historical hourly electricity demand

Country/region	Data source	Link
Albania	Open Power System Data	https://open-power-system-data.org/
Australia	Australian Energy Market Operator (AEMO)	https://aemo.com.au/
Austria	Open Power System Data	https://open-power-system-data.org/
Belgium	Open Power System Data	https://open-power-system-data.org/
Bosnia and Herzegovina	Open Power System Data	https://open-power-system-data.org/
Brazil	Operador Nacional do Sistema Elétrico (ONS)	https://www.gov.br/mme/pt-br
Bulgaria	Open Power System Data	https://open-power-system-data.org/
Canada	Ontario: Independent Electricity System Operator (IESO)	https://ieso.ca/
	Alberta: Alberta Electric System Operator (AESO)	https://www.aeso.ca/
Chile	Coordinador Electrico Nacional	https://www.coordinador.cl/
China	National Development and Reform Commission	https://www.ndrc.gov.cn/xxgk/zcfb/tz/201912/P020191230336066090861.pdf
Croatia	Open Power System Data	https://open-power-system-data.org/
Cyprus	Open Power System Data	https://open-power-system-data.org/
Czech Republic	Open Power System Data	https://open-power-system-data.org/
Denmark	Open Power System Data	https://open-power-system-data.org/
Estonia	Open Power System Data	https://open-power-system-data.org/
Finland	Open Power System Data	https://open-power-system-data.org/
France	Open Power System Data	https://open-power-system-data.org/
Germany	Open Power System Data	https://open-power-system-data.org/
Greece	Open Power System Data	https://open-power-system-data.org/
Hungary	Open Power System Data	https://open-power-system-data.org/
Iceland	Open Power System Data	https://open-power-system-data.org/
Ireland	Open Power System Data	https://open-power-system-data.org/
Italy	Open Power System Data	https://open-power-system-data.org/
Japan	TEPCO Tokyo Electric Power Company	https://www.tepco.co.jp/en/hd/index-e.html
Latvia	Open Power System Data	https://open-power-system-data.org/
Lithuania	Open Power System Data	https://open-power-system-data.org/
Luxembourg	Open Power System Data	https://open-power-system-data.org/
Macedonia	Open Power System Data	https://open-power-system-data.org/
Montenegro	Open Power System Data	https://open-power-system-data.org/
Netherlands	Open Power System Data	https://open-power-system-data.org/
New Zealand	Electricity Authority	https://www.ea.govt.nz/
Norway	Open Power System Data	https://open-power-system-data.org/
Poland	Open Power System Data	https://open-power-system-data.org/

Country/region	Data source	Link
Portugal	Open Power System Data	https://open-power-system-data.org/
Romania	Open Power System Data	https://open-power-system-data.org/
Russia	Russia's Federal Grid Company	https://www.fsk-ees.ru/
Serbia	Open Power System Data	https://open-power-system-data.org/
Singapore	Singapore's Energy Market Authority	https://www.ema.gov.sg/
Slovakia	Open Power System Data	https://open-power-system-data.org/
Slovenia	Open Power System Data	https://open-power-system-data.org/
Spain	Open Power System Data	https://open-power-system-data.org/
Sweden	Open Power System Data	https://open-power-system-data.org/
Switzerland	Open Power System Data	https://open-power-system-data.org/
Turkey	Open Power System Data	https://open-power-system-data.org/
Ukraine	Open Power System Data	https://open-power-system-data.org/
UK	Open Power System Data	https://open-power-system-data.org/
U.S.	U.S. Electric System Operating Data	https://toolkit.climate.gov/tool/us-electric-system-operating-data

Supplementary Table 11 | Cost assumptions for different technologies in the near current period

Type	Technology	Cost*	Unit	Reference
Overnight capital costs	Gas	813 (254–1,109)	\$/kW	IEA, 2020 ⁴¹
	Coal	1,871 (800–2,582)	\$/kW	
	Oil	1,342 (680–1,820)	\$/kW	
	Nuclear	3,944 (2,157–6,920)	\$/kW	
	Solar	1,000 (579–2,006)	\$/kW	
	Wind	1,387 (877–2,282)	\$/kW	NREL,2023 ⁴²
	Hydro	3,430 (1,899–5,891)	\$/kW	
	CCUS	5,896	\$/kW	
	Short-duration and long-duration	415.5 and 135	\$/kWh	USDOE, 2019 ⁴³
Variable costs	Gas	46 (24–69)	\$/MWh	IEA, 2020 ⁴¹
	Coal	41 (35–48)	\$/MWh	
	Oil	43 (32–58)	\$/MWh	
	Nuclear	26 (15–40)	\$/MWh	
Variable costs	Solar	1.06E-08	\$/MWh	Caldeira, 2019 ⁴⁴
	Wind	1.01E-08	\$/MWh	
	Hydro	0	\$/MWh	Zhuo et al., 2022 ⁴⁵
	CCUS	93	\$/MWh	NREL,2023 ⁴²
	Short-duration and long-duration	0	\$/MWh	Duan et al., 2022 ⁴⁶

*These costs vary across different countries, but only the mean and ranges are shown in this table due to limited space.

Supplementary Table 12 | Cost and technology assumptions of AC and DC transmission lines in 7 key countries in the near-current periods*

Country	Regions	UHV DC installed capacity	UHV AC installed capacity	UHV DC capital cost	UHV AC capital cost	UHV DC variable cost	UHV AC variable cost	UHV DC line loss	UHV AC line loss
		MW	MW	\$/MW	\$/MW	\$/MWh	\$/MWh	%	%
U.S.	SE	0	29,570	164,995	27,731	0	0	1.6	9.4
	SW	400	19,230	164,995	27,731	0	0	1.6	9.4
	TEN	0	26,470	164,995	27,731	0	0	1.6	9.4
	CAL	1,920	19,180	164,995	27,731	0	0	1.6	9.4
	NW	2,990	9,340	164,995	27,731	0	0	1.6	9.4
	NE	0	4,500	164,995	27,731	0	0	1.6	9.4
	MIDA	0	48,060	164,995	27,731	0	0	1.6	9.4
	FLA	0	6,650	164,995	27,731	0	0	1.6	9.4
	MIDW	200	62,640	164,995	27,731	0	0	1.6	9.4
	CENT	820	24,780	164,995	27,731	0	0	1.6	9.4
	NY	0	11,110	164,995	27,731	0	0	1.6	9.4
	CAR	0	15,750	164,995	27,731	0	0	1.6	9.4
	TEX	0	820	164,995	27,731	0	0	1.6	9.4
	Northeast	0	4,150	164,995	27,731	0	0	1.6	9.4
Brazil	North	0	4,650	164,995	27,731	0	0	1.6	9.4
	Southeast/Midwest	3,200	4,800	164,995	27,731	0	0	1.6	9.4
	South	1,800	3,300	164,995	27,731	0	0	1.6	9.4
	Central	0	121,200	0	101,343	0	0	0	1.5
	East	123,900	216,000	192,238	101,161	0	0	4.9	1.9
China (Mainland)	North	0	36,000	0	162,843	0	0	0	2.2
	Northeast	0	45,600	0	57,871	0	0	0	1.5
	Northwest	39,669	50,700	245,386	74,233	0	0	7.6	1.3
	South	146,970	42,000	416,003	124,543	0	0	8.6	2.3
	Southwest	177,300	57,900	227,790	120,529	0	0	4.3	1.8

Country	Regions	UHV DC installed capacity	UHV AC installed capacity	UHV DC capital cost	UHV AC capital cost	UHV DC variable cost	UHV AC variable cost	UHV DC line loss	UHV AC line loss
		MW	MW	\$/MW	\$/MW	\$/MWh	\$/MWh	%	%
India	South	6,000	5,390	141,176	64,205	0	0	1.6	3.4
	Northeast	0	2,400	141,176	48,190	0	0	0	2.5
	East	0	11,580	141,176	58,373	0	0	0	3.3
	Central	6,000	21,455	141,176	43,983	0	0	1.6	3.3
	West	6,000	9,180	141,176	78,387	0	0	1.6	3.3
	North	6,000	7,650	141,176	43,643	0	0	1.6	3.3
Germany	TransnetBW	0	6,200	164,995	27,730	0	0	0	9.4
	TeneT	0	17,440	164,995	27,730	0	0	0	9.4
	50Hertz	0	4,400	164,995	27,730	0	0	0	9.4
	Amprion	0	15,880	164,995	27,730	0	0	0	9.4
South Africa	East	533	5,860	161,760	27,187	0	0	1.6	3.0
	Central	0	7,060	161,760	27,187	0	0	0	2.8
	North	0	7,060	161,760	27,187	0	0	0	3.0
	Southwest	80	4,260	161,760	27,187	0	0	1.6	3.2
	NSW	180	1,166	161,760	27,187	0	0	1.6	6.0
Australia	Northern Territory	0	0	161,760	27,187	0	0	0	6.0
	QLD	0	506	161,760	27,187	0	0	0	6.0
	SA	80	275	161,760	27,187	0	0	1.6	6.0
	TAS	0	132	161,760	27,187	0	0	0	6.0
	VIC	720	1,067	161,760	27,187	0	0	1.6	6.0
	Western	0	0	161,760	27,187	0	0	0	6.0

* We assume a threefold expansion of transmission capacity relative to the near-present period ⁴⁷.

Supplementary Table 13 | Cost and technology assumptions of AC and DC transmission lines in European countries in the near-current periods*

Regions	UHV DC installed capacity	UHV AC installed capacity	UHV DC capital cost	UHV AC capital cost	UHV DC variable cost	UHV AC variable cost	UHV DC line loss	UHV AC line loss
	MW	MW	\$/MW	\$/MW	\$/MWh	\$/MWh	%	%
Albania	0	2,480	164,995	27,731	0	0	0	9.4
Austria	0	15,720	164,995	27,731	0	0	0	9.4
Belgium	838	4,320	164,995	27,731	0	0	1.6	9.4
Bosnia and Herzegovina	0	8,740	164,995	27,731	0	0	0	9.4
Bulgaria	0	4,880	164,995	27,731	0	0	0	9.4
Croatia	0	11,220	164,995	27,731	0	0	0	9.4
Cyprus	0	0	164,995	27,731	0	0	0	9.4
Czech Republic	0	7,940	164,995	27,731	0	0	0	9.4
Denmark	2,933	1,240	164,995	27,731	0	0	1.6	9.4
Estonia	838	1,580	164,995	27,731	0	0	1.6	9.4
Finland	1,676	1,380	164,995	27,731	0	0	1.6	9.4
France	4,609	12,160	164,995	27,731	0	0	1.6	9.4
Germany	1,676	23,120	164,995	27,731	0	0	1.6	9.4
Greece	419	2,720	164,995	27,731	0	0	1.6	9.4
Hungary	0	8,100	164,995	27,731	0	0	0	9.4
Iceland	0	0	164,995	27,731	0	0	0	9.4
Ireland	419	0	164,995	27,731	0	0	1.6	9.4
Italy	3,352	9,060	164,995	27,731	0	0	1.6	9.4
Kosovo	0	1,880	164,995	27,731	0	0	0	9.4
Latvia	0	3,760	164,995	27,731	0	0	0	9.4
Lithuania	419	3,180	164,995	27,731	0	0	1.6	9.4
Luxembourg	0	2,280	164,995	27,731	0	0	0	9.4
Macedonia	0	2,940	164,995	27,731	0	0	0	9.4
Moldova	4,680	1,380	164,995	27,731	0	0	1.6	9.4

Regions	UHV DC installed capacity	UHV AC installed capacity	UHV DC capital cost	UHV AC capital cost	UHV DC variable cost	UHV AC variable cost	UHV DC line loss	UHV AC line loss
	MW	MW	\$/MW	\$/MW	\$/MWh	\$/MWh	%	%
Montenegro	419	4,060	164,995	27,731	0	0	1.6	9.4
Netherlands	838	5,200	164,995	27,731	0	0	1.6	9.4
Norway	2,933	3,200	164,995	27,731	0	0	1.6	9.4
Poland	419	6,240	164,995	27,731	0	0	1.6	9.4
Portugal	0	4,580	164,995	27,731	0	0	0	9.4
Romania	0	6,040	164,995	27,731	0	0	0	9.4
Serbia	0	6,240	164,995	27,731	0	0	0	9.4
Slovakia	0	6,260	164,995	27,731	0	0	0	9.4
Slovenia	0	6,160	164,995	27,731	0	0	0	9.4
Spain	838	6,560	164,995	27,731	0	0	1.6	9.4
Sweden	3,352	3,820	164,995	27,731	0	0	1.6	9.4
Switzerland	0	15,500	164,995	27,731	0	0	0	9.4
Ukraine	0	1,380	164,995	27,731	0	0	0	9.4
UK	3,352	400	164,995	27,731	0	0	1.6	9.4

* We assume a threefold expansion of transmission capacity relative to the near-present period⁴⁷.

Supplementary Table 14 | Fixed and variable costs across key 7 countries in the near-current periods

Factors*	U.S.	Brazil	China	India	Germany	South Africa	Australia	Unit
<i>inv_nuclear</i>	4,250	3,606.5	2,500	2,778.0	5,466.5	3,606.5	3,606.5	\$/kW
<i>inv_hydro</i>	5,891.0	2,326.0	4,134.0	2,449.0	3,000	3,356.6	3,356.6	\$/kW
<i>inv_coal</i>	2,582.0	2,189.0	800	1,148.0	1,817.4	1,817.4	2,433.0	\$/kW
<i>inv_oil</i>	1,767.0	1,573.5	680	1,036.7	1,211.7	1,326.2	1,694.0	\$/kW
<i>inv_gas</i>	952.0	958.0	560	925.3	606.0	834.9	955.0	\$/kW
<i>inv_wind</i>	1,319.0	1,314.0	1,174.0	877.0	1,298.5	1,376.1	1,376.1	\$/kW
<i>inv_solar</i>	1,072.0	1,197.0	730	629.0	823.0	977.1	882.0	\$/kW
<i>inv_stor1</i>	415.5	415.5	415.5	415.5	415.5	415.5	415.5	\$/kWh
<i>inv_stor2</i>	135.0	135.0	135.0	135.0	135.0	135.0	135.0	\$/kWh
<i>OMF_nuclear</i>	85.0	72.1	50	55.6	109.3	72.1	72.1	\$/kW/year
<i>OMF_hydro</i>	117.8	46.5	82.7	49.0	60	67.1	67.1	\$/kW/year
<i>OMF_coal</i>	51.6	43.8	16.0	23.0	36.3	36.3	48.7	\$/kW/year
<i>OMF_oil</i>	35.3	31.5	13.6	20.7	24.2	26.5	33.9	\$/kW/year
<i>OMF_gas</i>	19.0	19.2	11.2	18.5	12.1	16.7	19.1	\$/kW/year
<i>OMF_wind</i>	39.6	39.4	35.2	26.3	39.0	41.3	41.3	\$/kW/year
<i>OMF_solar</i>	10.7	12.0	7.3	6.3	8.2	9.8	8.8	\$/kW/year
<i>OMF_stor1</i>	8.3	8.3	8.3	8.3	8.3	8.3	8.3	\$/kWh/year
<i>OMF_stor2</i>	2.7	2.7	2.7	2.7	2.7	2.7	2.7	\$/kWh/year
<i>vari_nuclear</i>	20.9	27.0	36.4	33.2	21.3	27.0	27.0	\$/MWh
<i>vari_hydro</i>	0	0	0	0	0	0	0	\$/MWh
<i>vari_coal</i>	42.6	40.4	40.3	40.9	40.9	40.9	38.3	\$/MWh
<i>vari_oil</i>	39.9	35.2	35.2	41.1	41.1	43.5	50.5	\$/MWh
<i>vari_gas</i>	23.7	30	67	41.2	41.2	46.0	62.6	\$/MWh
<i>vari_wind</i>	0	0	0	0	0	0	0	\$/MWh
<i>vari_solar</i>	0	0	0	0	0	0	0	\$/MWh
<i>vari_stor1</i>	0	0	0	0	0	0	0	\$/MWh
<i>vari_stor2</i>	0	0	0	0	0	0	0	\$/MWh

**inv*, *OMF*, and *vari* indicate overnight capital cost, fixed O&M cost, and variable cost, respectively. *stor1* and *stor2* denote short-duration and long-duration storage, respectively.

Supplementary Table 15 | Fixed and variable costs across 38 European countries in the near-current periods

Factors*	Albania	Austria	Belgium	BIH**	Bulgaria	Croatia	Cyprus	Unit
inv_nuclear	5,466.5	5,466.5	5,466.5	5,466.5	5,466.5	5,466.5	5,466.5	\$/kW
inv_hydro	2,592	2,361	2,592	2,592	2,592	2,592	2,592	\$/kW
inv_coal	1,817.4	1,817.4	1,817.4	1,817.4	1,817.4	1,817.4	1,817.4	\$/kW
inv_oil	1,211.7	1,211.7	1,395.7	1,211.7	1,211.7	1,211.7	1,211.7	\$/kW
inv_gas	606	606	974	606	606	606	606	\$/kW
inv_wind	1,298.5	1,458	1,475	1,298.5	1,298.5	1,298.5	1,298.5	\$/kW
inv_solar	823	719	992	823	823	823	823	\$/kW
inv_stor1	415.5	415.5	415.5	415.5	415.5	415.5	415.5	\$/kWh
inv_stor2	135	135	135	135	135	135	135	\$/kWh
OMF_nuclear	109.3	109.3	109.3	109.3	109.3	109.3	109.3	\$/kW/year
OMF_hydro	51.8	47.2	51.8	51.8	51.8	51.8	51.8	\$/kW/year
OMF_coal	36.3	36.3	36.3	36.3	36.3	36.3	36.3	\$/kW/year
OMF_oil	24.2	24.2	27.9	24.2	24.2	24.2	24.2	\$/kW/year
OMF_gas	12.1	12.1	19.5	12.1	12.1	12.1	12.1	\$/kW/year
OMF_wind	39	43.7	44.3	39	39	39	39	\$/kW/year
OMF_solar	8.2	7.2	9.9	8.2	8.2	8.2	8.2	\$/kW/year
OMF_stor1	8.3	8.3	8.3	8.3	8.3	8.3	8.3	\$/kWh/year
OMF_stor2	2.7	2.7	2.7	2.7	2.7	2.7	2.7	\$/kWh/year
vari_nuclear	21.3	21.3	21.3	21.3	21.3	21.3	21.3	\$/MWh
vari_hydro	0	0	0	0	0	0	0	\$/MWh
vari_coal	40.9	40.9	40.9	40.9	40.9	42.6	42.6	\$/MWh
vari_oil	41.1	41.1	44.2	41.1	41.1	35.9	35.9	\$/MWh
vari_gas	41.2	41.2	47.5	41.2	41.2	29.1	29.1	\$/MWh
vari_wind	0	0	0	0	0	0	0	\$/MWh
vari_solar	0	0	0	0	0	0	0	\$/MWh
vari_stor1	0	0	0	0	0	0	0	\$/MWh
vari_stor2	0	0	0	0	0	0	0	\$/MWh

[illegible]

[illegible]

Factors*	LUX**	Macedonia	Moldova	MNE**	Netherlands	Norway	Poland	Unit
inv_nuclear	5,466.5	5,466.5	5,466.5	5,466.5	5,466.5	5,466.5	5,466.5	\$/kW
inv_hydro	2,592	2,592	2,592	2,592	2,592	1,899	2,592	\$/kW
inv_coal	1,817.4	1,817.4	1,817.4	1,817.4	1,817.4	1,817.4	1,817.4	\$/kW
inv_oil	1,211.7	1,211.7	1,211.7	1,211.7	1,211.7	1,211.7	1,211.7	\$/kW
inv_gas	606	606	606	606	606	606	606	\$/kW
inv_wind	1,298.5	1,298.5	1,298.5	1,298.5	1,250	930	1,298.5	\$/kW
inv_solar	823	823	823	823	807	984	823	\$/kW
inv_stor1	415.5	415.5	415.5	415.5	415.5	415.5	415.5	\$/kWh
inv_stor2	135	135	135	135	135	135	135	\$/kWh
OMF_nuclear	109.3	109.3	109.3	109.3	109.3	109.3	109.3	\$/kW/year
OMF_hydro	51.8	51.8	51.8	51.8	51.8	38	51.8	\$/kW/year
OMF_coal	36.3	36.3	36.3	36.3	36.3	36.3	36.3	\$/kW/year
OMF_oil	24.2	24.2	24.2	24.2	24.2	24.2	24.2	\$/kW/year
OMF_gas	12.1	12.1	12.1	12.1	12.1	12.1	12.1	\$/kW/year
OMF_wind	39	39	39	39	37.5	27.9	39	\$/kW/year
OMF_solar	8.2	8.2	8.2	8.2	8.1	9.8	8.2	\$/kW/year
OMF_stor1	8.3	8.3	8.3	8.3	8.3	8.3	8.3	\$/kWh/year
OMF_stor2	2.7	2.7	2.7	2.7	2.7	2.7	2.7	\$/kWh/year
vari_nuclear	21.3	21.3	21.3	21.3	21.3	21.3	21.3	\$/MWh
vari_hydro	0	0	0	0	0	0	0	\$/MWh
vari_coal	40.9	40.9	40.9	40.9	40.9	40.9	40.9	\$/MWh
vari_oil	41.1	41.1	41.1	41.1	41.1	41.1	41.1	\$/MWh
vari_gas	41.2	41.2	41.2	41.2	41.2	41.2	41.2	\$/MWh
vari_wind	0	0	0	0	0	0	0	\$/MWh
vari_solar	0	0	0	0	0	0	0	\$/MWh
vari_stor1	0	0	0	0	0	0	0	\$/MWh
vari_stor2	0	0	0	0	0	0	0	\$/MWh

[illegible]

Factors*	Switzerland	Ukraine	UK	Unit
<i>inv_nuclear</i>	5,466.5	5,466.5	5,466.5	\$/kW
<i>inv_hydro</i>	2,592	2,592	2,592	\$/kW
<i>inv_coal</i>	1,817.4	1,817.4	1,817.4	\$/kW
<i>inv_oil</i>	1,211.7	1,211.7	1,211.7	\$/kW
<i>inv_gas</i>	606	606	606	\$/kW
<i>inv_wind</i>	1,298.5	1,298.5	1,298.5	\$/kW
<i>inv_solar</i>	823	823	823	\$/kW
<i>inv_stor1</i>	415.5	415.5	415.5	\$/kWh
<i>inv_stor2</i>	135	135	135	\$/kWh
<i>OMF_nuclear</i>	109.3	109.3	109.3	\$/kW/year
<i>OMF_hydro</i>	51.8	51.8	51.8	\$/kW/year
<i>OMF_coal</i>	36.3	36.3	36.3	\$/kW/year
<i>OMF_oil</i>	24.2	24.2	24.2	\$/kW/year
<i>OMF_gas</i>	12.1	12.1	12.1	\$/kW/year
<i>OMF_wind</i>	39	39	39	\$/kW/year
<i>OMF_solar</i>	8.2	8.2	8.2	\$/kW/year
<i>OMF_stor1</i>	8.3	8.3	8.3	\$/kWh/year
<i>OMF_stor2</i>	2.7	2.7	2.7	\$/kWh/year
<i>vari_nuclear</i>	21.3	21.3	21.3	\$/MWh
<i>vari_hydro</i>	0	0	0	\$/MWh
<i>vari_coal</i>	40.9	40.9	40.9	\$/MWh
<i>vari_oil</i>	41.1	41.1	41.1	\$/MWh
<i>vari_gas</i>	41.2	41.2	41.2	\$/MWh
<i>vari_wind</i>	0	0	0	\$/MWh
<i>vari_solar</i>	0	0	0	\$/MWh
<i>vari_stor1</i>	0	0	0	\$/MWh
<i>vari_stor2</i>	0	0	0	\$/MWh

**inv*, *OMF*, and *vari* indicate overnight capital cost, fixed O&M cost, and variable cost, respectively. *stor1* and *stor2* denote short-duration and long-duration storage, respectively. **CZE, BIH, LUX, and MNE indicate Czech Republic, Bosnia and Herzegovina, Luxembourg, and Montenegro, respectively.

Supplementary Table 16 | Decision variables in the dispatch optimization model

Symbol	Unit	Description	Classification
Cap_i	MW	Install capacity for i category generation technology	Base model
Cap_s	MW	Install capacity for s category storage technology	Base model
$Gen_{i,t}$	MWh	Power generation for i category generation technology at hour t	Base model
$Stor_{s,t}^{cha}$	MWh	Charging electricity for s category storage system at hour t	Base model
$Stor_{s,t}^{dis}$	MWh	Discharging electricity at hour t for s category storage system	Base model
$SOC_{s,t}$	MWh	State of charge for s category storage energy state at hour t	Base model
$G_{a,t}^{g,k}$	MWh	AC transmission electricity from region g to region k at hour t	Nested model
$G_{d,t}^{g,k}$	MWh	DC transmission electricity from region g to region k at hour t	Nested model
Cap_i^g	MW	Install capacity for i category generation technology at region g	Nested model
Cap_s^g	MW	Install capacity for s category storage technology at region g	Nested model
$G_{i,t}^g$	MWh	Power generation for i category generation technology at hour t at region g	Nested model
$Stor_{s,t}^{cha,g}$	MWh	Charging electricity for s category storage system at hour t at region g	Nested model
$Stor_{s,t}^{dis,g}$	MWh	Discharging electricity for s category storage system at hour t at region g	Nested model
$SOC_{s,t}^g$	MWh	State of charge for s category storage energy state at hour t at region g	Nested model
ΔD_t	MWh	Changed electricity demand at hour t	Demand response
Imp_t	MWh	Imported electricity at hour t	Expanded interconnection

Supplementary Table 17 | Power generation structures of 26 major countries in 2020*

Country	Share (%)								
	Bioenergy	Coal	Gas	Hydro	Nuclear	Solar	Wind	Other fossils (mainly oil)	Other renewables
Algeria	0	0	98.62	0.06	0	0.85	0.01	0.45	0
Argentina	1.52	1.13	61.34	16.36	6.92	0.93	6.51	5.29	0
Australia	1.29	53.91	20.01	5.44	0	8.99	8.53	1.84	0
Botswana	0	97.25	0	0	0	0.46	0	2.29	0
Brazil	9.34	3.69	9.08	63.99	2.47	1.71	9.07	2.32	0
China	1.74	63.22	3.18	16.99	4.71	3.36	6	0.69	0
Egypt	0	0	85.22	7.50	0	2.24	2.63	2.4	0
Finland	16.71	3.52	5.76	22.93	33.64	0.32	11.93	5.18	0
France	1.66	0.65	6.79	11.65	67.11	2.5	7.56	1.9	0.12
Germany	8.99	23.57	16.07	3.25	11.26	8.73	23.31	3.75	0.04
India	2.08	72.09	4.54	10.48	2.86	3.71	3.82	0.2	0
Japan	3	29.74	35.18	7.71	4.28	7.51	0.83	2.5	0
Kazakhstan	0.01	63.68	24.51	8.89	0	1.14	0.95	0.82	0
Luxembourg	30.33	0	14.75	7.38	0	13.11	28.69	5.74	0
Mexico	2.15	6.03	58.46	8.56	3.64	4.86	6.05	6.34	0
Namibia	0	3.00	0	75.00	0	20.5	1	0.5	0
New Zealand	1.8	3.92	13.70	55.43	0	0.37	5.26	1.45	18.06
Nigeria	0.06	0	77.90	21.93	0	0.11	0	0	0
Paraguay	0.3	0	0	99.70	0	0	0	0	0
Russia	0.07	14.03	44.73	19.57	19.89	0.17	0.11	1.14	0
South Africa	0.17	84.51	0.79	0.21	6.51	2.38	2.68	0.62	0
South Korea	2.43	36.32	26.71	0.68	27.91	3.35	0.55	1.25	0
Tajikistan	0	8.26	1.64	90.10	0	0	0	0	0
U.S.	1.35	19.69	40.55	6.73	19.40	3.23	8.36	0.85	0.45
UK	12.55	1.73	36.48	2.08	16.08	3.97	24.01	3.8	0
Uruguay	20.12	0	0	30.14	0	3.39	40.38	5.97	0

*This data is collected from <https://ember-energy.org/data/>

Supplementary Table 18 | The proportion of different power generation technologies for in European countries*

Region	Nuclear (%)	Hydro (%)	Oil (%)	Gas (%)	Coal (%)
Albania	0	99.44	0	0	0
Austria	0	60.83	0.81	4.50	14.41
Belgium	39.00	0.31	0.09	4.01	30.34
Bosnia and Herzegovina	0	27.28	70.58	0.24	0
Bulgaria	41.43	7.03	33.66	0.60	5.71
Croatia	0	42.78	9.22	0.15	26.00
Cyprus	0	0	0	87.63	0
Czech Republic	37.50	2.67	38.70	2.31	8.58
Denmark	0	0.07	10.71	3.11	4.13
Estonia	0	0.49	0	50.66	0.49
Finland	33.64	22.93	3.52	5.18	5.76
France	67.11	11.65	1.97	6.79	0.65
Germany	11.26	3.25	0.75	16.07	23.57
Greece	0	7.03	13.91	8.67	40.48
Hungary	45.99	0.69	10.62	1.55	26.03
Iceland	0	68.40	0	0	0
Ireland	0	2.91	2.10	5.01	50.81
Italy	0	17.18	4.83	4.63	48.30
Kosovo	0	3.88	94.48	0.15	0
Latvia	0	45.45	0	0	36.36
Lithuania	0	6.38	0	9.15	36.17
Luxembourg	0	7.38	0	5.74	14.75
Macedonia	0	23.97	49.25	1.69	21.35
Moldova	0	4.53	0	0	94.17
Montenegro	0	42.77	47.79	0	0
Netherlands	3.35	0.04	6.25	4.10	59.46
Norway	0	91.89	0.03	0.41	1.04
Poland	0	1.35	68.34	2.69	11.00
Portugal	0	23.51	4.59	2.57	34.25
Romania	20.78	27.86	16.95	0.56	17.13
Serbia	0	24.25	71.07	0.13	1.37
Slovakia	54.16	15.85	6.52	2.77	12.52
Slovenia	37.60	29.19	25.81	0.12	3.43
Spain	22.47	11.76	2.13	4.67	26.88
Sweden	30.04	44.20	0	1.43	0.06
Switzerland	35.07	55.29	0	1.83	0.91
Ukraine	51.56	5.12	27.20	0.17	9.60
UK	16.08	2.08	1.73	0.29	36.48

*This data is collected from <https://ember-energy.org/data/>

Supplementary Table 19 | Cost and technology assumptions for four individual strategies

Measure	Factor	Sub-factor	Assumption			Reference
Demand response	Peak response coefficient	Base	NA			Billings and Powell, 2023 ⁶ ; Debon Securities, 2023 ⁷ ; FERC, 2010 ⁴⁸ ; FERC, 2022 ⁴⁹ ; NDRC, 2023 ⁵⁰ ; Yuan, 2020 ⁵¹
		Aggressive	7%			
		Intermediate	5%			
		Conservative	3%			
	Cost assumption	10% variable cost of thermal power			Topsperity Securities, 2023 ⁵²	
Expanded interconnection	Proportion of imported power	Base	NA			EIA, 2025 ⁸ ; EMBER, 2025 ⁹
		Aggressive	10%			
		Intermediate	5%			
		Conservative	3%			
	Cost assumption	Average hourly costs × 2				
Enhanced storage	Discharge/charge efficiency	Short duration storage		Long duration storage		Chen et al., 2021 ¹⁰ ; Jenkins and Sepulveda, 2021 ⁵³ ; Staadecker et al., 2024 ⁵⁴ ; Yang et al., 2023 ¹¹ ; Yang et al., 2024 ⁵⁵
		Base	91%	80%		
		Aggressive	99.50%	99%		
		Intermediate	99%	95%		
		Conservative	97%	93%		
	Cost assumption	Fixed cost × 1.05				
	Improved thermal	Minimal output	Coal		Oil	Gas
Base			50%	40%	30%	
Aggressive			25%	20%	15%	
Intermediate			30%	25%	20%	
Conservative			35%	30%	25%	
Cost assumption		Fixed cost × 1.20				

Supplementary Table 20 | Net import rate of 26 major countries in 2022*

Region	Country	Electricity demand (TWh)	Net import (TWh)	Net import rate (%)
America	U.S.	4,328.1	41.21	0.95
	Brazil	690.08	12.91	1.87
	Mexico	343.25	3.1	0.9
	Argentina	158.5	12.88	8.13
	Uruguay	13.08	-2.1	-16.06
	Paraguay	22.29	-21.72	-97.44
Asia	China	8,834.4	-14.31	-0.16
	India	1,828.85	-0.45	-0.02
	Japan	1,040.57	0	0
	South Korea	624.6	0	0
	Kazakhstan	112.63	-0.82	-0.73
	Tajikistan	18.71	-2.17	-11.6
Europe	Russia	1,149.89	-17.05	-1.48
	Germany	538.97	-27.26	-5.06
	France	484.28	14.94	3.08
	UK	317.99	-4.3	-1.35
	Finland	84.12	12.52	14.88
	Luxembourg	6.69	5.51	82.36
Africa	Egypt	215.28	-1.43	-0.66
	South Africa	235.08	-3.64	-1.55
	Nigeria	36.98	0	0
	Algeria	87.04	-1.14	-1.31
	Botswana	4.29	1.66	38.69
	Namibia	3.8	2.45	64.47
Oceania	Australia	272.92	0	0
	New Zealand	44.19	0	0

*This data is collected from <https://ember-energy.org/data/yearly-electricity-data/>

Supplementary Table 21 | Details of technology-specific generation costs for 26 major countries in the near current periods

Country	Technology	Cost (\$/kWh)	Reference and link
U.S.	Bioenergy	0.094	https://www.desn.com.cn/news/show-1804405.html
U.S.	Coal	0.114	
U.S.	Gas	0.058	Projected Costs of Generating Electricity 2020, https://www.oecd-neo.org/jcms/pl_51110
U.S.	Hydro	0.101	
U.S.	Nuclear	0.047	
U.S.	Solar	0.062	https://ourworldindata.org/grapher/levelized-cost-of-energy
U.S.	Wind	0.034	
Brazil	Bioenergy	0.054	https://www.oecd-neo.org/lcoe/
Brazil	Coal	0.097	
Brazil	Gas	0.065	Projected Costs of Generating Electricity 2020, https://www.oecd-neo.org/jcms/pl_51110
Brazil	Hydro	0.046	
Brazil	Solar	0.064	https://ourworldindata.org/grapher/levelized-cost-of-energy
Brazil	Wind	0.027	
Mexico	Bioenergy	0.076	
Mexico	Coal	0.08	Analysis of levelised cost of electricity generation and penetration rates of grid-connected solar PV and on-shore wind technologies, https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20190725082445910
Mexico	Gas	0.043	Projected Costs of Generating Electricity 2020, https://www.oecd-neo.org/jcms/pl_51110
Mexico	Solar	0.062	https://ourworldindata.org/grapher/levelized-cost-of-energy
Mexico	Wind	0.047	
Argentina	Coal	0.093	Analysis of levelised cost of electricity generation and penetration rates of grid-connected solar PV and on-shore wind technologies, https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20190725082445910
Argentina	Gas	0.043	Refer to the gas power costs in Mexico. Projected Costs of Generating Electricity 2020, https://www.oecd-neo.org/jcms/pl_51110
Argentina	Solar	0.052	https://ourworldindata.org/grapher/levelized-cost-of-energy
Argentina	Wind	0.072	
China	Bioenergy	0.06	https://www.cpn.com.cn/news/gj/202108/t20210827_1423545.html
China	Coal	0.050	https://www.sohu.com/a/335608966_100038373
China	Gas	0.084	Projected Costs of Generating Electricity 2020, https://www.oecd-neo.org/jcms/pl_51110
China	Hydro	0.039	Sustainable small-scale hydropower solutions in Central Asian countries for local and cross-border energy/water supply, https://doi.org/10.1016/j.rser.2022.112726
China	Nuclear	0.066	Projected Costs of Generating Electricity 2020, https://www.oecd-neo.org/jcms/pl_51110
China	Solar	0.038	https://ourworldindata.org/grapher/levelized-cost-of-energy
China	Wind	0.03	
India	Bioenergy	0.118	https://www.oecd-neo.org/lcoe

Country	Technology	Cost (\$/kWh)	Reference and link
India	Coal	0.055	Analysis of levelised cost of electricity generation and penetration rates of grid-connected solar PV and on-shore wind technologies, https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20190725082445910
India	Gas	0.074	
India	Hydro	0.049	Projected Costs of Generating Electricity 2020, https://www.oecd-neo.org/jcms/pl_51110
India	Nuclear	0.066	
India	Solar	0.041	https://ourworldindata.org/grapher/levelized-cost-of-energy
India	Wind	0.04	
Japan	Bioenergy	0.076	Renewable Power Generation Costs in 2020, https://www.irena.org/publications/2021/Jun/Renewable-Power-Costs-in-2020
Japan	Coal	0.1	
Japan	Gas	0.093	Projected Costs of Generating Electricity 2020, https://www.oecd-neo.org/jcms/pl_51110
Japan	Hydro	0.142	
Japan	Nuclear	0.087	
Japan	Solar	0.104	https://ourworldindata.org/grapher/levelized-cost-of-energy
Japan	Wind	0.136	
South Korea	Bioenergy	0.076	Renewable Power Generation Costs in 2020, https://www.irena.org/publications/2021/Jun/Renewable-Power-Costs-in-2020
South Korea	Coal	0.076	
South Korea	Gas	0.091	Projected Costs of Generating Electricity 2020, https://www.oecd-neo.org/jcms/pl_51110
South Korea	Hydro	0.053	
South Korea	Solar	0.068	https://ourworldindata.org/grapher/levelized-cost-of-energy
South Korea	Wind	0.073	
Tajikistan	Coal	0.072	Analysis of levelised cost of electricity generation and penetration rates of grid-connected solar PV and on-shore wind technologies, https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20190725082445910
Tajikistan	Hydro	0.060	https://assets.publishing.service.gov.uk/media/57a0897b40f0b652dd00023e/61646_Levelised-Cost-of-Electricity.pdf
Russia	Coal	0.055	
Russia	Nuclear	0.042	Projected Costs of Generating Electricity 2020, https://www.oecd-neo.org/jcms/pl_51110
Russia	Solar	0.052	https://ourworldindata.org/grapher/levelized-cost-of-energy
Russia	Wind	0.064	
Germany	Coal	0.076	https://china-neo.cn/site/content/22799.html
Germany	Hydro	0.057	Projected Costs of Generating Electricity 2020, https://www.oecd-neo.org/jcms/pl_51110
Germany	Solar	0.069	
Germany	Wind	0.053	https://ourworldindata.org/grapher/levelized-cost-of-energy
France	Solar	0.058	
France	Wind	0.049	
UK	Solar	0.077	

Country	Technology	Cost (\$/kWh)	Reference and link
UK	Wind	0.04	https://ourworldindata.org/grapher/levelized-cost-of-energy
Egypt	Gas	0.049	Analysis of levelised cost of electricity generation and penetration rates of grid-connected solar PV and on-shore wind technologies, https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20190725082445910
Egypt	Solar	0.052	https://ourworldindata.org/grapher/levelized-cost-of-energy
Egypt	Wind	0.049	https://ourworldindata.org/grapher/levelized-cost-of-energy
South Africa	Coal	0.080	https://assets.publishing.service.gov.uk/media/57a0897b40f0b652dd00023e/61646_Levelised-Cost-of-Electricity.pdf
South Africa	Gas	0.080	https://assets.publishing.service.gov.uk/media/57a0897b40f0b652dd00023e/61646_Levelised-Cost-of-Electricity.pdf
South Africa	Solar	0.067	https://ourworldindata.org/grapher/levelized-cost-of-energy
South Africa	Wind	0.049	https://ourworldindata.org/grapher/levelized-cost-of-energy
Nigeria	Coal	0.055	Analysis of levelised cost of electricity generation and penetration rates of grid-connected solar PV and on-shore wind technologies,
Nigeria	Gas	0.067	https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20190725082445910
Nigeria	Wind	0.057	Projected Costs of Generating Electricity 2020, https://www.oecd-neo.org/jcms/pl_51110
Botswana	Coal	0.087	Refer to the coal power costs in neighboring Zimbabwe. https://assets.publishing.service.gov.uk/media/57a0897b40f0b652dd00023e/61646_Levelised-Cost-of-Electricity.pdf
Botswana	Wind	0.057	Projected Costs of Generating Electricity 2020, https://www.oecd-neo.org/jcms/pl_51110
Australia	Bioenergy	0.01	The roles of biomass and CSP in a 100 % renewable electricity supply in Australia
Australia	Coal	0.103	Projected Costs of Generating Electricity 2020, https://www.oecd-neo.org/jcms/pl_51110
Australia	Gas	0.112	https://www.oecd-neo.org/jcms/pl_51110
Australia	Solar	0.046	https://ourworldindata.org/grapher/levelized-cost-of-energy
Australia	Wind	0.04	https://ourworldindata.org/grapher/levelized-cost-of-energy
New Zealand	Bioenergy	0.076	Renewable Power Generation Costs in 2020, https://www.irena.org/publications/2021/Jun/Renewable-Power-Costs-in-2020
New Zealand	Coal	0.091	https://www.mbie.govt.nz/building-and-energy/energy-and-natural-resources/energy-statistics-and-modelling/energy-modelling/interactive-levelised-cost-of-electricity-comparison-tool
New Zealand	Gas	0.082	Projected Costs of Generating Electricity 2020, https://www.oecd-neo.org/jcms/pl_51110
New Zealand	Hydro	0.055	https://www.oecd-neo.org/jcms/pl_51110
New Zealand	Solar	0.052	https://ourworldindata.org/grapher/levelized-cost-of-energy
New Zealand	Wind	0.037	https://ourworldindata.org/grapher/levelized-cost-of-energy
European Union	Bioenergy	0.087	https://www.cpn.com.cn/news/gj/202108/t20210827_1423545.html
European Union	Coal	0.13	
European Union	Gas	0.09	World Energy Outlook 2020, https://www.iea.org/reports/world-energy-outlook-2020
European Union	Hydro	0.064	
European Union	Nuclear	0.15	

Country	Technology	Cost (\$/kWh)	Reference and link
European Union	Solar	0.055	Analysis of levelised cost of electricity generation and penetration rates of grid-connected solar PV and on-shore wind technologies, https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20190725082445910 Renewable Power Generation Costs in 2020, https://www.irena.org/publications/2021/Jun/Renewable-Power-Costs-in-2020 Analysis of levelised cost of electricity generation and penetration rates of grid-connected solar PV and on-shore wind technologies, https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20190725082445910 Analysis of levelised cost of electricity generation and penetration rates of grid-connected solar PV and on-shore wind technologies, https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20190725082445910 Renewable Power Generation Costs in 2020, https://www.irena.org/publications/2021/Jun/Renewable-Power-Costs-in-2020 https://www.cpn.com.cn/news/gj/202108/t20210827_1423545.html Renewable Power Generation Costs in 2020, https://www.irena.org/publications/2021/Jun/Renewable-Power-Costs-in-2020 Analysis of levelised cost of electricity generation and penetration rates of grid-connected solar PV and on-shore wind technologies, https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20190725082445910 Renewable Power Generation Costs in 2020, https://www.irena.org/publications/2021/Jun/Renewable-Power-Costs-in-2020 Analysis of levelised cost of electricity generation and penetration rates of grid-connected solar PV and on-shore wind technologies, https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20190725082445910
European Union	Wind	0.055	
Africa	Coal	0.052	
Africa	Gas	0.065	
Africa	Hydro	0.083	
Oceania	Hydro	0.145	
Asia	Coal	0.057	
Asia	Gas	0.065	
Asia	Hydro	0.092	
North America	Bioenergy	0.097	
North America	Hydro	0.112	Analysis of levelised cost of electricity generation and penetration rates of grid-connected solar PV and on-shore wind technologies, https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20190725082445910 Renewable Power Generation Costs in 2020, https://www.irena.org/publications/2021/Jun/Renewable-Power-Costs-in-2020 Analysis of levelised cost of electricity generation and penetration rates of grid-connected solar PV and on-shore wind technologies, https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20190725082445910 Renewable Power Generation Costs in 2020, https://www.irena.org/publications/2021/Jun/Renewable-Power-Costs-in-2020 Analysis of levelised cost of electricity generation and penetration rates of grid-connected solar PV and on-shore wind technologies, https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20190725082445910 Renewable Power Generation Costs in 2020, https://www.irena.org/publications/2021/Jun/Renewable-Power-Costs-in-2020 Analysis of levelised cost of electricity generation and penetration rates of grid-connected solar PV and on-shore wind technologies, https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20190725082445910 Renewable Power Generation Costs in 2020, https://www.irena.org/publications/2021/Jun/Renewable-Power-Costs-in-2020 Analysis of levelised cost of electricity generation and penetration rates of grid-connected solar PV and on-shore wind technologies, https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20190725082445910
Central America and the Caribbean	Coal	0.08	
Central America and the Caribbean	Gas	0.074	
Central America and the Caribbean	Hydro	0.052	
Other South America	Hydro	0.079	
Middle East	Gas	0.066	
World	Bioenergy	0.076	
World	Coal	0.055	
World	Gas	0.067	
World	Hydro	0.044	
World	Nuclear	0.057	Renewable Power Generation Costs in 2020, https://www.irena.org/publications/2021/Jun/Renewable-Power-Costs-in-2020
World	Solar	0.039	
World	Wind	0.057	
World	Other Fossil	0.037	
World	Other Renewables	0.071	

Supplementary Table 22 | IAM-predicted wind and solar share under the SSP126 scenario by 2060

Country	10th percentile	Mean	90th percentile
Algeria	40.45%	70.47%	80.81%
Argentina	25.28%	56.07%	72.83%
Australia	29.93%	59.75%	81.61%
Botswana	44.72%	65.68%	86.44%
Brazil	22.98%	38.01%	51.26%
China	29.03%	60.76%	80.80%
Egypt	40.45%	70.47%	80.81%
Finland	44.05%	63.80%	86.63%
France	44.05%	63.80%	86.63%
Germany	44.05%	63.80%	86.63%
India	48.55%	69.42%	89.12%
Japan	16.60%	44.16%	66.99%
Kazakhstan	17.58%	50.83%	75.40%
Luxembourg	44.05%	63.80%	86.63%
Mexico	24.92%	59.69%	79.60%
Namibia	44.72%	65.68%	86.44%
New Zealand	33.33%	51.53%	67.98%
Nigeria	53.25%	59.83%	64.28%
Paraguay	33.83%	56.07%	73.32%
Russia	11.77%	47.17%	75.26%
South Africa	47.18%	67.17%	89.84%
South Korea	11.88%	46.13%	79.77%
Tajikistan	17.58%	50.83%	75.40%
UK	44.05%	63.80%	86.63%
U.S.	45.91%	63.94%	89.25%
Uruguay	33.83%	56.07%	73.32%

Supplementary Table 23 | IAM-projected wind/solar generation ratio by 2060 under the SSP126 scenario

Country	Wind power	Solar power
Algeria	41.9%	58.1%
Argentina	56.3%	43.7%
Australia	50.1%	49.9%
Botswana	49.8%	50.2%
Brazil	66.5%	33.5%
China	57.1%	42.9%
Egypt	41.9%	58.1%
Finland	63.8%	36.2%
France	63.8%	36.2%
Germany	63.8%	36.2%
India	45.5%	54.5%
Japan	67.3%	32.7%
Kazakhstan	58.2%	41.8%
Luxembourg	63.8%	36.2%
Mexico	51.1%	48.9%
Namibia	49.8%	50.2%
New Zealand	66.9%	33.1%
Nigeria	70.0%	30.0%
Paraguay	56.3%	43.7%
Russia	59.4%	40.6%
South Africa	60.2%	39.8%
South Korea	50.8%	49.2%
Tajikistan	58.2%	41.8%
UK	63.8%	36.2%
U.S.	57.4%	42.6%
Uruguay	56.3%	43.7%

Supplementary Table 24 | IAM-predicted generation proportion of CCUS to thermal power by 2060 under the SSP126 scenario

Country	CCUS proportion
Algeria	30.76%
Argentina	74.97%
Australia	61.50%
Botswana	65.99%
Brazil	29.20%
China	71.19%
Egypt	99.84%
Finland	64.81%
France	78.71%
Germany	78.71%
India	62.73%
Japan	85.21%
Kazakhstan	73.60%
Luxembourg	78.71%
Mexico	66.17%
Namibia	64.81%
New Zealand	81.56%
Nigeria	45.32%
Paraguay	67.74%
Russia	53.08%
South Africa	81.50%
South Korea	75.41%
Tajikistan	73.60%
UK	78.71%
U.S.	64.85%
Uruguay	64.81%

Supplementary Table 25 | IAM-projected 10 percentile of thermal power generation by 2060 under the SSP126 scenario

Country	Thermal power share
Algeria	14.15%
Argentina	2.44%
Australia	5.80%
Botswana	0.58%
Brazil	0.00%
China	0.09%
Egypt	11.02%
Finland	0.07%
France	0.07%
Germany	0.07%
India	0.33%
Japan	4.86%
Kazakhstan	0.00%
Luxembourg	0.07%
Mexico	1.02%
Namibia	0.58%
New Zealand	1.26%
Nigeria	1.96%
Paraguay	0.03%
Russia	0.00%
South Africa	0.00%
South Korea	4.48%
Tajikistan	0.00%
UK	0.07%
U.S.	0.08%
Uruguay	0.03%

Supplementary Table 26 | Data sources of subnational grid interconnections for 7 key countries

Country	Factor	Data source
Mainland China	Hourly electricity demand	https://www.ndrc.gov.cn/xxgk/zcfb/tz/201912/P020191230336066090861.pdf
	Power grid SHP file	
	UHV DC installed capacity	
	UHV AC installed capacity	
	UHV DC capital cost	
	UHV AC capital cost	
	UHV DC operational cost	Zhuo et al., 2022 ⁴⁵
	UHV AC operational cost	
	UHV DC line loss	
	UHV AC line loss	
	Sectoral power cost	
	Sectoral generation share	https://www.ndrc.gov.cn/xxgk/zcfb/tz/201912/P020191230336066090861.pdf
India	Hourly electricity demand	
	Power grid SHP file	https://iced.niti.gov.in/
	UHV DC installed capacity	
	UHV AC installed capacity	https://iced.niti.gov.in/
	UHV DC capital cost	https://www.nature.com/articles/s41467-020-18318-7
	UHV AC capital cost	Sasse and Trutnevyte, 2020 ⁵⁶
	UHV DC operational cost	
	UHV AC operational cost	
	UHV DC line loss	Bogdanov et al., 2019 ⁵⁷
	UHV AC line loss	
	Sectoral power cost	IEA, 2020 ⁴¹

Country	Factor	Data source
Germany	Sectoral generation share	https://iced.niti.gov.in/
	Hourly electricity demand	https://transparency.entsoe.eu/load-domain/r2/totalLoadR2/show
	Power grid SHP file	https://www.netztransparenz.de/de-de/
	UHV DC installed capacity	https://transparency.entsoe.eu/load-domain/r2/totalLoadR2/show
	UHV AC installed capacity	
	UHV DC capital cost	Sasse and Trutnevyte, 2020 ⁵⁶
	UHV AC capital cost	
	UHV DC operational cost	
	UHV AC operational cost	Bogdanov et al., 2019 ⁵⁷
	UHV DC line loss	
	UHV AC line loss	IEA, 2020 ⁴¹
	Sectoral power cost	
South Africa	Sectoral generation share	https://transparency.entsoe.eu/
	Hourly electricity demand	https://www.datafirst.uct.ac.za/dataportal/index.php/catalog/?page=1&sort_by=title&sort_order=asc&ps=15 and https://figshare.com/articles/dataset/PSO-BP_codes_for_modelling_the_relationships_between_the_national_GDP_and_nighttime_light_data/19517272/1
	Power grid SHP file	https://www.datafirst.uct.ac.za/dataportal/index.php/catalog/?page=1&sort_by=title&sort_order=asc&ps=15
	UHV DC installed capacity	https://www.mdpi.com/1996-1073/15/12/4356
	UHV AC installed capacity	https://www.eskom.co.za/dataportal/
	UHV DC capital cost	Sasse and Trutnevyte, 2020 ⁵⁶
	UHV AC capital cost	
	UHV DC operational cost	
	UHV AC operational cost	Bogdanov et al., 2019 ⁵⁷
	UHV DC line loss	
	UHV AC line loss	IEA, 2020 ⁴¹
	Sectoral power cost	

Country	Factor	Data source
U.S.	Sectoral generation share	https://www.statista.com/statistics/1237610/south-africa-distribution-of-electricity-production-by-source/
	Hourly electricity demand	https://www.eia.gov/electricity/gridmonitor/dashboard/custom/pending
	Power grid SHP file	
	UHV DC installed capacity	https://www.nrel.gov/grid/national-transmission-planning-study.html
	UHV AC installed capacity	
	UHV DC capital cost	Sasse and Trutnevyte, 2020 ⁵⁶
	UHV AC capital cost	
	UHV DC operational cost	
	UHV AC operational cost	Bogdanov et al., 2019 ⁵⁷
	UHV DC line loss	
	UHV AC line loss	
	Sectoral power cost	IEA, 2020 ⁴¹
	Sectoral generation share	https://www.eia.gov/electricity/gridmonitor/about
Australia	Hourly electricity demand	https://www.aemo.com.au/
	Power grid SHP file	
	UHV DC installed capacity	https://www.aemo.com.au/
	UHV AC installed capacity	
	UHV DC capital cost	Sasse and Trutnevyte, 2020 ⁵⁶
	UHV AC capital cost	
	UHV DC operational cost	
	UHV AC operational cost	Bogdanov et al., 2019 ⁵⁷
	UHV DC line loss	
	UHV AC line loss	
	Sectoral power cost	IEA, 2020 ⁴¹
	Sectoral generation share	https://www.aemo.com.au/

Country	Factor	Data source
Brazil	Hourly electricity demand	https://www.ons.org.br/paginas/resultados-da-operacao/historico-da-operacao/dados-gerais
	Power grid SHP file	https://app.electricitymaps.com/zone/BR-S/all
	UHV DC installed capacity	https://ri.taesa.com.br/en/taesa/transmission-sector/
	UHV AC installed capacity	
	UHV DC capital cost	Sasse and Trutnevyte, 2020 ⁵⁶
	UHV AC capital cost	
	UHV DC operational cost	
	UHV AC operational cost	Bogdanov et al., 2019 ⁵⁷
	UHV DC line loss	
	UHV AC line loss	
	Sectoral power cost	IEA, 2020 ⁴¹
	Sectoral generation share	https://www.ons.org.br/paginas/resultados-da-operacao/historico-da-operacao/dados-gerais

Supplementary Table 27 | The proportion of different generation technologies for each subnational grid in 7 key countries in the near current periods

Country	Subnational grid	Nuclear (%)	Hydro (%)	Oil (%)	Gas (%)	Coal (%)
U.S.	SE	20.5	4.75	0	51.9	19.3
	SW	28	3.69	0	42.1	16.2
	TEN	41.4	12.1	0	29.6	16.1
	CAL	8.33	11.4	0	47.4	2.82
	NW	2.52	33.7	0	21.7	24
	NE	27.2	7.59	0	54.7	0.35
	MIDA	33.3	1.98	0	39.4	19.5
	FLA	12.2	0.09	0	68.7	8.04
	MIDW	14	1.63	0	32.7	37.9
	CENT	5.55	5.57	0	24	31.1
	NY	25	21.4	0	44.4	0
	CAR	44.2	2.05	0	26.3	19.4
	TEX	9.87	0.13	0	44.6	17.4
Brazil	Northeast	0	32.14	0.63	3.59	0.72
	North	0	85.95	1.58	9.01	1.81
	Southeast/Midwest	4.95	81.37	1.58	9.03	1.82
	South	0	82.11	1.46	8.31	1.67
China (Mainland)	Central	0	17.78	0.02	0.46	60.81
	East	11.64	4.31	0.02	6.08	63.79
	North	1.01	0.79	0.04	7.78	67.35
	Northeast	12.25	5.10	0.08	0.11	56.07
	Northwest	0	12.60	0	0.32	62.08
	South	12.79	21.66	0	8.82	45.46
	Southwest	0	63.36	0	2.40	26.71
India	South	5.90	6.03	0.25	0.25	65.10
	Northeast	0	40.65	11.43	11.43	34.26
	East	0	5.42	0	0	68.32
	Central	0.58	1.70	0.16	0.16	93.65
	West	5.15	3.59	2.00	2.00	68.82
	North	2.73	29.38	0.37	0.37	46.78
Germany	TransnetBW	18.19	20.87	0.31	3.93	27.03
	TeneT	7.59	6.61	0.03	7.18	9.80
	50Hertz	0	5.41	1.78	6.62	41.41
	Amprion	6.78	4.21	0	13.46	48.15
South Africa	East	4.23	1.3	0	0	84.4
	Central	4.23	1.3	0	0	84.4
	North	4.23	1.3	0	0	84.4
	Southwest	4.23	1.3	0	0	84.4
Australia	NSW	0	0.09	43.70	9.50	37.10
	Northern Territory	0	0.17	40.48	22.45	21.15
	QLD	0	0.09	39.10	18.10	33.40
	SA	0	0.10	36.60	19.90	32.30
	TAS	0	0.17	49.20	25.40	7.50
	VIC	0	0.41	39.70	7.30	8.90
	Western	0	0.17	34.60	54.50	7.70

Supplementary Table 28 | Data source of international grid interconnections for European countries

Factor	Data source
Power grid SHP file	https://www.entsoe.eu/data/map/
UHV DC installed capacity	https://www.entsoe.eu/data/map/
UHV AC installed capacity	https://www.entsoe.eu/data/map/
UHV DC capital cost	Sasse and Trutnevyte, 2020 ⁵⁶
UHV AC capital cost	
UHV DC operational cost	
UHV AC operational cost	Bogdanov et al., 2019 ⁵⁷
UHV DC line loss	
UHV AC line loss	

Supplementary references

1. Parzen, M. *et al.* PyPSA-Earth. A new global open energy system optimization model demonstrated in Africa. *Appl. Energy* **341**, 121096. <https://doi.org/10.1016/j.apenergy.2023.121096> (2023).
2. Nahid, F.A., Roy, J. PyPSA-BD: A customized model to explore decarbonized energy transition for developing country. *Renew. Energy Focus* **52**, 100655. <https://doi.org/10.1016/j.ref.2024.100655> (2025).
3. Tong, D. *et al.* Geophysical constraints on the reliability of solar and wind power worldwide. *Nat. Commun* **12**, 6146. <https://doi.org/10.1038/s41467-021-26355-z> (2021).
4. Liu, L. *et al.* Climate change impacts on planned supply–demand match in global wind and solar energy systems. *Nat. Energy* **8**, 870–880. <https://doi.org/10.1038/s41560-023-01304-w> (2023).
5. Staffell, I., Pfenninger, S. & Johnson, N. A global model of hourly space heating and cooling demand at multiple spatial scales. *Nat. Energy* **8**, 1328–1344. <https://doi.org/10.1038/s41560-023-01341-5> (2023).
6. Billings, B. W., Powell, K. M. Are regulations enough to expand industrial demand response? A study of the impacts of policy on industrial demand response in the United States. *Electr. J* **36**, 107270. <https://doi.org/10.1016/j.tej.2023.107270> (2023).
7. Debon Securities. *Langxin Technology: Dual-track energy digitization + energy internet, assisting the development of new power systems (in Chinese)*. <https://www.hangyan.co/reports/3220331677628761251> (2023).
8. EIA. *International energy consumption*. <https://www.eia.gov/international/overview/world> (2025).
9. EMBER. *Yearly Electricity Data – Yearly electricity data – long format (CSV)*, <https://ember-energy.org/data/yearly-electricity-data/> (2025).
10. Chen, X. *et al.* Pathway toward carbon-neutral electrical systems in China by mid-century with negative CO₂ abatement costs informed by high-resolution modeling. *Joule* **5**, 2715–2741. <https://doi.org/10.1016/j.joule.2021.10.006> (2021).
11. Yang, G. *et al.* Implications of future price trends and interannual resource uncertainty on firm solar power delivery with photovoltaic overbuilding and battery storage. *IEEE Trans. Sustain. Energy* **14**, 2036–2048. <https://doi.org/10.1109/TSTE.2023.3274109> (2023).
12. Li, W., Sun, W., Li, B., Han, Y. & He, Y. Flexibility transformation strategy of thermal power units for typical scenario considering wind and solar consumption based on generalized Benders decomposition algorithm. *Electr. Eng.* **105**, 3423–3432. <https://doi.org/10.1007/s00202-023-01890-8> (2023).
13. Pang, Y., Chi, Y. & Tian, B. Economic evaluation of flexible transformation in coal-fired power plants with multi price links. *J. Clean. Prod.* **402**, 136851. <https://doi.org/10.1016/j.jclepro.2023.136851> (2023).

14. Sherman, P., Chen, X. & McElroy, M. B. Wind-generated electricity in China: Decreasing potential, inter-annual variability and association with changing climate. *Sci Rep* **7**, 16294. <https://doi.org/10.1038/s41598-017-16073-2> (2017).
15. Karinauskas, K. B., Lundquist, J. K. & Zhang, L. Southward shift of the global wind energy resource under high carbon dioxide emissions. *Nat. Geosci* **11**, 38-43. <https://doi.org/10.1038/s41561-017-0029-9> (2018).
16. Zeng, Z. *et al.* A reversal in global terrestrial stilling and its implications for wind energy production. *Nat. Clim. Change* **9**, 979-985. <https://doi.org/10.1038/s41558-019-0622-6> (2019).
17. Gao, M. *et al.* Secular decrease of wind power potential in India associated with warming in the Indian Ocean. *Sci. Adv.* **4**, eaat5256. <https://doi.org/10.1126/sciadv.aat5256> (2018).
18. Pryor, S.C., Barthelmie, R.J., Bukovsky, M.S., Leung, L.R. & Sakagushi K. Climate change impacts on wind power generation. *Nat. Rev Earth Environ* **1**, 627–643. <https://doi.org/10.1038/s43017-020-0101-7> (2020).
19. Rantanen, M. *et al.* The Arctic has warmed nearly four times faster than the globe since 1979. *Commun Earth Environ* **3**, 168. <https://doi.org/10.1038/s43247-022-00498-3> (2022).
20. Toktarova, A., Gruber, L., Hlusiak, M., Bogdanov, D. & Breyer, C. Long term load projection in high resolution for all countries globally. *Int. J. Electr. Power Energy Syst.* **111**, 160-181. <https://doi.org/10.1016/j.ijepes.2019.03.055> (2019).
21. Hersbach, H. *et al.* The ERA5 global reanalysis. *Q. J. R. Meteorol. Soc.* **146**, 1999-2049. <https://doi.org/10.1002/qj.3803> (2020).
22. IRENA (International Renewable Energy Agency). *Future of solar photovoltaic: Deployment, investment, technology, grid integration and socio-economic aspects*. <https://resources.solarbusinesshub.com/images/reports/228.pdf> (2019).
23. ACER, *ACER Market Monitoring Report 2015 - Electricity and gas retail markets*, Agency for the Cooperation of Energy Regulators, Ljubljana, Slovenia. https://www.acer.europa.eu/sites/default/files/documents/en/Electricity/Market%20monitoring/MMR_2015_Corrigenda/MMR%202015-RETAIL_V2.pdf (2016).
24. Prokhorov, O. & Dreisbach, D. The impact of renewables on the incidents of negative prices in the energy spot markets. *Energy Policy* **167**, 113073. <https://doi.org/10.1016/j.enpol.2022.113073> (2022).
25. European Commission. *Electricity price statistics*. https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Electricity_price_statistics (accessed 17 April 2025).
26. IRENA, *Innovation landscape for a renewable-powered future: Solutions to integrate variable renewables*. Summary for policy makers. International Renewable Energy Agency, Abu Dhabi. https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2019/Feb/IRENA_Innovation_Landscape_2019_report.pdf (2019).

27. Enrich, J., Li, R., Mizrahi, A., Reguant, M. Measuring the impact of time-of-use pricing on electricity consumption: Evidence from Spain. *J. Environ. Econ. Manage.* **123**, 102901. <https://doi.org/10.1016/j.jeem.2023.102901> (2024).
28. ENTSO-E. *ENTSO-E Transparency Platform*. <https://newtransparency.entsoe.eu/market> (accessed 17 April 2025).
29. EPEX SPOT. *Basics of the Power Market*. <https://www.epexspot.com/en/basicspowermarket> (accessed 17 April 2025).
30. Cabot, C. & Villavicencio, M. Second-best electricity pricing in France: Effectiveness of existing rates in evolving power markets. *Energy Econ.* **136**, 107673. <https://doi.org/10.1016/j.eneco.2024.107673> (2024).
31. Xin, X. *et al.* *IPCC DDC: BCC BCC-CSM2MR model output prepared for CMIP6 ScenarioMIP*. <https://doi.org/10.26050/WDCC/AR6.C6SPBCBCM> (World Data Center for Climate (WDCC) at DKRZ, 2023).
32. Swart, N.C. *et al.* *CCCma CanESM5 model output prepared for CMIP6 ScenarioMIP*. *Earth System Grid Federation*. <https://doi.org/10.22033/ESGF/CMIP6.1317> (2019).
33. Shiogama, H., Abe, M. & Tatebe, H. *MIROC MIROC6 model output prepared for CMIP6 ScenarioMIP*. <https://doi.org/10.22033/ESGF/CMIP6.898> (Earth System Grid Federation, 2019).
34. Watanabe, S. *et al.* *MIROC MIROC-ES2H model output prepared for CMIP6 GeoMIP*. <https://doi.org/10.22033/ESGF/CMIP6.907> (Earth System Grid Federation, 2021).
35. von Storch, J.-S. *et al.* *MPI-M MPIESM1.2-HR model output prepared for CMIP6 HighResMIP*. <https://doi.org/10.22033/ESGF/CMIP6.762> (Earth System Grid Federation, 2017).
36. Yukimoto, S. *et al.* *IPCC DDC: MRI MRI-ESM2.0 model output prepared for CMIP6 CMIP*. <https://doi.org/10.26050/WDCC/AR6.C6CMMRME0> (World Data Center for Climate (WDCC) at DKRZ, 2023).
37. Cao, J. *IPCC DDC: NUIST NESMv3 model output prepared for CMIP6 ScenarioMIP*. World Data Center for Climate (WDCC) at DKRZ. <https://doi.org/10.26050/WDCC/AR6.C6SPNUIINES> (2023).
38. ESA. *Land Cover CCI Product User Guide Version 2*. https://maps.elie.ucl.ac.be/CCI/viewer/download/ESACCI-LC-Ph2-PUGv2_2.0.pdf (2017).
39. Danielson, J. J. & Gesch, D. B. *Global multi-resolution terrain elevation data 2010 (GMTED2010)*. Report No. 2011-1073. (2011).
40. UNEP-WCMC and IUCN, *Protected Planet: The World Database on Protected Areas (WDPA) and World Database on Other Effective Area-based Conservation Measures (WD-OECM)* [Online], May 2025, Cambridge, UK: UNEP-WCMC and IUCN. Available at: www.protectedplanet.net (2025).

41. IEA. *Projected Costs of Generating Electricity* 2020. <https://www.iea.org/reports/projected-costs-of-generating-electricity-2020> (2020).
42. NREL (National Renewable Energy Laboratory). *Electricity Annual Technology Baseline (ATB) Data*. <https://atb.nrel.gov/electricity/2023/2023/data> (2023).
43. U.S. DOE (The Pacific Northwest National Laboratory of the U.S.). *Energy Storage Technology and Cost Characterization Report*. https://www.energy.gov/sites/prod/files/2019/07/f65/Storage%20Cost%20and%20Performance%20Characterization%20Report_Final.pdf (2019).
44. Caldeira, K. *Simple Energy Model (SEM-1)*, <https://github.com/ClabEnergyProject/SEM-1.1> (2019).
45. Zhuo, Z. *et al.* Cost increase in the electricity supply to achieve carbon neutrality in China. *Nat. Commun* **13**, 3172. <https://doi.org/10.1038/s41467-022-30747-0> (2022).
46. Duan, L., Petroski, R., Wood, L. & Caldeira, K. Stylized least-cost analysis of flexible nuclear power in deeply decarbonized electricity systems considering wind and solar resources worldwide. *Nat Energy* **7**, 260–269. <https://doi.org/10.1038/s41560-022-00979-x> (2022).
47. Fan, J.L. *et al.* A net-zero emissions strategy for China's power sector using carbon-capture utilization and storage. *Nat Commun* **14**, 5972. <https://doi.org/10.1038/s41467-023-41548-4> (2023).
48. FERC (Federal Energy Regulatory Commission). *National action plan on demand response*. https://www.energy.gov/sites/prod/files/oeprod/DocumentsandMedia/FERC_NAPDR_-_final.pdf (2010).
49. FERC (Federal Energy Regulatory Commission). *2022 assessment of demand response and advanced metering*. <https://www.ferc.gov/media/2022-assessment-demand-response-and-advanced-metering> (2022).
50. NDRC (National Development and Reform Commission). *Electricity demand side management measures* (2023 Edition, in Chinese). <https://www.ndrc.gov.cn/xxgk/zcfb/ghxwj/202309/P020230927316131533276.pdf> (2023).
51. Yuan, J. *Economic analysis and security pathways for China's power supply security* (In Chinese). http://doc.cserver.com.cn/doc_1b2aec83-b7cf-46ab-861d-38f67e33a746.html (2020).
52. Topsperity Securities. *Langxin Technology: The dual-track approach of energy digitization and energy internet development propels the construction of new power systems, ushering in opportunities for rapid advancement* (in Chinese). <https://www.hangyan.co/reports/3220331677628761251> (2023).
53. Jenkins, J. D. & Sepulveda, N. A. Long-duration energy storage: A blueprint for research and innovation. *Joule* **5**, 2241–2246. <https://doi.org/10.1016/j.joule.2021.08.002> (2021).

54. Staadecker, M., Szinai, J., Sánchez-Pérez, P. A., Kurtz, S. & Hidalgo-Gonzalez, P. The value of long-duration energy storage under various grid conditions in a zero-emissions future. *Nat. Commun* **15**, 9501. <https://doi.org/10.1038/s41467-024-53274-6> (2024).
55. Yang, G., Yang, D., Liu, B. & Zhang, H. The role of short- and long-duration energy storage in reducing the cost of firm photovoltaic generation. *Appl. Energy* **374**, 123914. <https://doi.org/10.1016/j.apenergy.2024.123914> (2024).
56. Sasse, J.-P. & Trutnevyte, E. Regional impacts of electricity system transition in Central Europe until 2035. *Nat. Commun* **11**, 4972. <https://doi.org/10.1038/s41467-020-18812-y> (2020).
57. Bogdanov, D. *et al.* Radical transformation pathway towards sustainable electricity via evolutionary steps. *Nat Commun* **10**, 1077. <https://doi.org/10.1038/s41467-019-08855-1> (2019).